CAR-01 · CAREER ROADMAP

The project controls career roadmap

The six disciplines, how they interlock, and the four transitions that decide a career

VERSION 1.0 · DRAFT

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A map of the project controls function as it is actually staffed: six interlocking disciplines, the artefacts each one owns, and the four transitions that separate someone who produces numbers from someone trusted to own them. It explains why almost nobody enters project controls deliberately, what that does to the shape of a career, and how to audit your own gaps against the function rather than against a job title.

About this document

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Contents

The project controls career roadmap

In one paragraph. Project controls is one function made of six disciplines — cost, planning and scheduling, risk, estimating, change and commercial, and reporting and analytics — which produce a single output: a defensible statement of where a project is and where it will end up. This document maps those disciplines, shows how one field event propagates through all six, and sets out the four transitions that decide a controls career: producing a number, owning it, integrating across accounts, and setting the control environment that produces numbers you never personally checked. It contains no pay, demand or market figures, and no claim about how long any rung takes; it says what must be demonstrated instead.

Who this is for. Cost engineers, planners, schedulers, risk analysts, estimators, quantity surveyors and project accountants who want a map of the whole function rather than of their own corner of it, and the managers who hire and develop them.

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1. The function before the career

A project controls function exists to answer four questions, on a fixed cycle, in a form that survives challenge:

1. What has this project cost so far, and is that number complete?
2. What will it cost when it is finished, and on what assumption?
3. When will it finish, and what is driving that date?
4. What could change either answer, how likely is it, and what have we set aside for it?

Everything else — the report pack, the dashboards, the tooling, the meetings — is apparatus for producing those four answers. A controls career is best understood as progressively larger custody of them.

That framing matters because job titles do not travel. Cost engineer, cost controller, cost analyst and project accountant describe overlapping work in different sectors and sometimes in different buildings of the same organisation. Planner, scheduler and planning engineer likewise. Reading a title tells you almost nothing; reading the artefacts a person owns tells you everything. The role taxonomy the Institute uses for levelling is set out in `SAL-03 — Role taxonomy and levelling`.

A note on what this series does not contain. There are no salary figures, pay ranges, demand statistics or hiring data anywhere in S07, and none should be inferred from it. Market questions are answered by an instrument, not by assertion: see `SAL-01 — Salary and skills report: framework and methodology` for how the Institute intends to gather that data, and `SAL-06 — Using market data honestly` for how to read anyone else's.

2. The six disciplines

Each discipline owns a question, a set of artefacts and a cadence. The artefacts are the point: they are what you can be held to, and later what you can evidence.

Discipline	The question it answers	Artefacts it owns	Typical cadence
Cost	What has it cost, and what will it cost?	Cost ledger reconciliation, commitment register, accrual schedule, control-account forecast, cash flow	Monthly, with a weekly commitment view
Planning and scheduling	When will it finish, and what is driving that?	Baseline programme, progress update, driving-path narrative, lookahead, schedule metrics	Weekly update, monthly report
Risk	What could change the answer, and what have we set aside?	Risk register, quantified analysis, contingency drawdown record, risk review minutes	Monthly review, event-driven update
Estimating	What should this cost before we commit to it?	Estimate basis, quantity take-off, rate build-up, benchmark file, estimate reconciliation	Front-end and per change
Change and commercial	What have we been instructed to do, and what may we claim?	Trend register, change order log, variation pricing, notices, entitlement position	Continuous, with a monthly position
Reporting and analytics	What does all of this mean, to whom, by when?	The report pack, the data model, the dashboard, the movement bridge	Monthly, against a published calendar

Three observations about that table that are worth more than the table itself.

The disciplines are not equally staffed. On smaller projects one person holds three or four of them. On a large capital programme each is a team. This is why career shape varies so widely: someone from a small-project background often has breadth and no depth, and someone from a large programme often has the reverse. Both are legitimate; both have a predictable gap.

Estimating is the discipline most often orphaned. It sits at the front end, frequently in a separate department, and its output becomes everyone else's baseline. Controls professionals who have never built an estimate tend to treat the budget as a fact rather than as a set of assumptions with a documented basis — and so cannot say which assumption has failed when the cost runs.

Reporting is not a discipline of presentation. It is the discipline of reconciliation: making the cost position, the schedule position, the commercial position and the risk position tell the same story about the same project on the same data date, and explaining any gap between them rather than smoothing it.

3. How they interlock

The interlock is not conceptual. It is a sequence of hand-offs that happens every month, and it fails at identifiable joints.

The month-end chain. Field quantities and timesheets produce progress. Progress plus rules of credit produces earned value. Invoices plus accruals produce actual cost. Earned value against actual cost produces performance indices. Performance indices plus judgement produce the forecast. The forecast against the budget produces the variance. The variance plus a narrative produces the report. Each arrow is a hand-off between people, and each hand-off has a characteristic failure: quantities claimed but not installed, accruals missed at cut-off, an index computed on mismatched data dates.

The change chain. An instruction arrives. Estimating prices it. Planning assesses its effect on the driving path. Commercial establishes entitlement and issues any notice within its time bar. Cost holds it as a pending change until approval, then moves it into the budget through baseline change control. Risk retires the corresponding entry from the register so the same exposure is not funded twice.

The risk chain. The register identifies exposure. Quantification prices it. The contingency held is justified by that quantification, not by a percentage. As risks are realised or retired, the drawdown record explains where contingency went — which is the only credible answer when someone asks why the forecast moved.

Section 7 walks one event through all three chains with arithmetic, because the interlock is easier to believe in numbers than in prose.

4. Why almost nobody arrives here deliberately

Project controls is usually entered sideways. People arrive from engineering because they were the one who kept the spreadsheet; from quantity surveying because measurement and valuation are adjacent to cost control; from finance because someone needed the ledger reconciled; from site supervision because they knew what had actually been installed; from data and analytics because the reporting had become a data problem. A minority study something explicitly labelled project controls and enter directly.

This has three consequences that shape every career in the function.

Gaps are idiosyncratic, not standard. Two people with the same job title can have entirely non-overlapping blind spots. The engineer who never learned cut-off discipline and the accountant who has never seen a rule of credit will both produce a monthly report, and the reports will fail differently. Self-assessment against a competency map is therefore not a bureaucratic exercise; it is the only reliable way to find out what you do not know. `CAR-02 — Routes into project controls` treats each entry route and its characteristic gap in detail.

Learning is by apprenticeship, and apprenticeship is inconsistent. What you know depends on who trained you and which project you were on when. Someone whose formative cycles were spent on a lump-sum contract with a contested delay has seen things that someone on a well-run reimbursable programme has not, and the reverse.

The standard therefore has to come from outside the employer. That is the Institute's reason for existing: to describe the discipline to one standard rather than leaving it to whichever organisation happened to train you. PCI is developed with reference to ISO/IEC 17024 personnel-certification principles and is not accredited; a credential evidences assessed knowledge against a published standard, and it does not promise a job, a promotion or a pay outcome.

5. The four transitions

Rungs are less useful than transitions, because a transition is a change in what you are trusted with, and that is what actually has to be demonstrated. Nothing here is expressed as a period of time: how long a transition takes depends on sector, employer, project size and how much of the work you are allowed anywhere near.

Transition 1 — from producing to owning. You stop assembling a number someone else stands behind and start choosing the assumption the number rests on. The visible evidence is that you can say what would change your mind and by when you will know. In cost this is the move from producing the report to owning the forecast, treated in `CAR-03`. In planning it is the move from maintaining the programme to defending it, treated in `CAR-04`.

Transition 2 — from one account to the whole position. You move from a set of control accounts to the project's total position, which means reconciling other people's work and finding the double count. The demonstrated capability is integration: the cost position, the programme, the risk provision and the commercial position tie to each other, and where they do not, you can explain why in one sentence.

Transition 3 — from reporting the position to being believed. The number is now challenged by people with an incentive to dislike it. What is demonstrated here is not accuracy but defensibility: a movement bridge from the last position, a named driver for every line, a stated test. This is the threshold of leadership and is the subject of `CAR-05 — Into project controls leadership`.

Transition 4 — from owning numbers to owning the system that produces them. You become accountable for positions you did not personally check, produced by people you hired, using a process you designed. The demonstrated capability is control design: the data model, the calendar, the change thresholds, the assurance regime, and a function that produces a defensible report when you are on leave.

Most stalled careers are stuck at transition 1 or transition 3. Transition 1 stalls because nobody hands over forecast ownership; you have to take it, in writing, before you are asked. Transition 3 stalls because being right is treated as sufficient, and it is not.

6. How this goes wrong

Collecting tools instead of transitions. A list of scheduling and cost systems on a CV describes what you have had access to, not what you have been trusted with. Software fluency is assumed at every rung and distinguishes nobody above the first.

Specialising before you can see the interlock. Deep cost skill with no idea how the programme is built produces forecasts that ignore the driver of the overrun. Deep planning skill with no cost literacy produces a critical path nobody funds. The strongest practitioners in the function are competent in one discipline and conversationally fluent in the other five.

Mistaking longevity for progression. Repeating the same monthly cycle many times over is experience of one cycle, not of a career. The test is whether the size and contestedness of the number you own has grown.

Waiting to be given the next transition. Forecast ownership, defence of a programme and control design are almost never handed over. They are taken by doing them alongside the current job — writing the forecast you were not asked for, keeping the delta narrative nobody requested — and being right often enough that the formal handover becomes an administrative step.

Letting the evidence evaporate. People leave a project with nothing but a job title. The decisions you made, the errors you caught and the movement you predicted are recoverable only while you are there, and only in a form you may lawfully keep. `CAR-07 — Building a portfolio of evidence` sets out how.

Treating the credential as the outcome. A credential evidences knowledge against a published standard. It does not evidence that you have owned a contested forecast, and no honest reading of it says otherwise.

7. Worked example — one event through six disciplines

Illustrative figures. Currency-neutral units. A single control account with a budget at completion of 2,000,000 units. A design change instructs an additional 400 m of cable containment and cable pull. Assumptions on which every number below depends: an installation rate of 3.2 labour-hours per metre; an all-in labour rate of 60 units per hour; a material rate of 45 units per metre; a crew of eight working nine hours per day; the activity holds 10 working days of total float at the data date; and the change is fully recoverable under the contract.

Estimating. Labour hours = 400 m × 3.2 h/m = 1,280 hours. Labour cost = 1,280 h × 60 = 76,800. Materials = 400 m × 45 = 18,000. Estimated change value = 76,800 + 18,000 = **94,800**.

Planning. Crew capacity = 8 people × 9 hours = 72 hours per working day. Duration = 1,280 ÷ 72 = 17.8 → **18 working days**. Against 10 days of total float, the effect on the driving path is 18 − 10 = **8 working days** of critical delay, before any recovery.

Change and commercial. The instruction is priced at 94,800 and notified within the contract's notice period. Until it is approved it is a pending change: it belongs in the forecast, not in the budget.

Cost. Budget at completion remains 2,000,000. Forecast final cost = 2,000,000 + 94,800 = **2,094,800**. The 94,800 gap is a pending-change exposure, not an overrun, and must be labelled as such in the report. On approval it moves into the budget through baseline change control and the gap closes to zero.

Risk. The register held "late design release on containment routing" at 40 % likelihood and 120,000 impact, an expected value of 0.40 × 120,000 = **48,000**. That exposure has now materialised as an instructed, funded variation. If the 48,000 stays in the risk provision while the 94,800 sits in the forecast, the same event is funded twice, and the project reports 48,000 of contingency it does not have.

Reporting. One line of the report must therefore carry, on the same data date: budget 2,000,000; approved change 0; pending change 94,800; forecast 2,094,800; risk provision reduced by 48,000; driving path moved out by 8 working days. Any report that shows the cost effect without the schedule effect, or the pending change without the risk retirement, is internally inconsistent — and someone will find it.

The teaching point is not the arithmetic, which is trivial. It is that six people produced those numbers and only one of them can see whether they agree.

8. Checklist — auditing yourself against the function

Work through this against the six disciplines in §2, not against your job description.

- For each of the six disciplines, name the artefact you have personally produced most recently, and the one you have never produced.
- For your own discipline, name the last decision you made that someone senior did not check.
- State your current forecast for something, in writing, with the assumption it rests on and the date by which you will know whether you were right. If you cannot, you are at transition 1.
- Take last month's report and trace one number backwards to its raw source — a timesheet, a measured quantity, an invoice. Note every hand-off where it could have been wrong.
- Identify one figure in your reporting that is produced by someone else and that you accept without a test. Design the test.
- Name the person in the commercial team who decides what you may claim, and the person in finance who owns the ledger you must reconcile to. If you cannot name both, your position is unverified.
- Write down the largest number you are trusted to be wrong about. That figure, not your title, is your current rung.

The consequence of skipping this audit is specific and common: you become excellent at the part of the cycle you inherited, invisible in the five parts you never touched, and surprised when the transition you were waiting for goes to someone who had already been doing it.

Related

- `CAR-02 — Routes into project controls` — the entry routes and the characteristic gap each one leaves
- `CAR-03 — Cost engineer to cost manager` — the cost ladder and the forecast-ownership transition
- `CAR-04 — Planner to planning manager` — the planning ladder and the transition from maintaining to defending
- `CAR-05 — Into project controls leadership` — running a function and being believed
- `CAR-07 — Building a portfolio of evidence` — how to capture what you have owned, lawfully
- `SAL-03 — Role taxonomy and levelling` — the Institute's role definitions behind this map

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 4 (performance management, variance analysis and management reporting), Domain 5 (cost management and cost control), Domain 6 (earned value management and forecasting), Domain 10 (project scheduling) and Domain 12 (risk management for project controls); and from the PCL-AI competency set as published by the Institute. No external market, salary, demand or hiring data has been used, and none should be inferred — see `SAL-01`.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

CAR-02 · CAREER ROADMAP

Routes into project controls

What each entry route genuinely transfers, and the gap it reliably leaves

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Project controls is staffed almost entirely by transfer, from engineering, quantity surveying and commercial, finance and accounting, site supervision, data and analytics, and directly from education. This document sets out, route by route, what genuinely transfers, the gap each route reliably leaves, the first three things to learn, the characteristic failure of an incomer, and the evidence that proves the transfer has been made.

About this document

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Contents

Routes into project controls

In one paragraph. Almost nobody enters project controls through a front door, so the useful question is not "how do I get in" but "what does my particular route leave me missing". This document takes the six routes people actually take — engineering, quantity surveying and commercial, finance and accounting, site supervision, data and analytics, and directly from education — and for each sets out what transfers unchanged, what has to be learned from scratch, the mistake that identifies an incomer within a fortnight, and the evidence that closes the gap. It also names the common core every route must eventually acquire, whatever it arrived with.

Who this is for. Engineers, quantity surveyors, accountants, supervisors, analysts and graduates considering a move into project controls, and the managers deciding which of them to take a chance on.

1. The function is staffed by transfer, and that is not an accident

A controls function needs someone who can read a contract, someone who knows what is physically installed, someone who can reconcile to a ledger, someone who can build a network of dependencies and someone who can model. No single degree produces that person, so organisations assemble the function out of adjacent disciplines and train the rest on the job.

The practical consequence for anyone moving in: you are being hired for the part you already have and tolerated for the part you do not, and the tolerance has a limit. Your first cycle is spent demonstrating the transferred skill; your first six cycles are spent quietly closing the gap. The people who stall are those who keep solving controls problems with the tool they arrived with — the engineer who models everything, the accountant who reconciles everything, the site supervisor who trusts the field over the ledger.

2. The common core, whatever route you took

Four things every route eventually has to acquire. They are the entry price to the function, not the distinguishing skill within it.

The data model. The work breakdown structure (WBS) divides scope; the cost breakdown structure (CBS) and the code of accounts divide money; the organisational breakdown structure assigns accountability. The control account is where they meet, and it is the unit of everything that follows. If you cannot say which control account a given invoice, timesheet or quantity belongs to, no amount of analytical skill will help you.

The period cycle. Cut-off, accrual, progress measurement, earned value, forecast, variance, report — in that order, to a published calendar, with a fixed data date. Most reporting errors are cycle errors: a cost from one period compared with progress from another.

The contract as the boundary. The contract decides what may be claimed, what must be notified and by when, what counts as a change, and what evidence is required. Controls that ignore the contract produce numbers that are internally consistent and commercially worthless.

Auditability. Every reported number must be traceable to a source that existed at the data date and has not been altered since. This is a habit, not a system, and it is the habit most incomers lack.

3. Route by route

Each route below is treated the same way: what transfers, what is missing, what to learn first, the characteristic failure, and what evidence proves the transfer has been made. Evidence here means the artefacts described in `CAR-07 — Building a portfolio of evidence`.

3.1 From engineering and design

Transfers. Quantities and how they are derived. Technical credibility with the delivery team — you can ask why an activity is late and understand the answer. A feel for design maturity, which is the single best early predictor of change. Comfort with structured decomposition, which makes the WBS obvious rather than arbitrary.

Missing. The commercial frame. Engineers are trained to optimise the technical answer and are rarely taught that the contract, not the design, decides what may be recovered. Also missing: cut-off discipline and the idea that a number's completeness matters more than its precision.

Learn first. Accrual and cut-off; the difference between commitment, accrual and actual cost; rules of credit for progress (see `BPG-06 — Progress measurement and rules of credit`).

Characteristic failure. Reporting technical truth as commercial truth — "the work is 70 % done" when the certified valuation is 55 % and the earned value 62 %, with no explanation of why those three numbers differ.

Evidence of transfer. An accrual position you built at cut-off that survived audit; a change you priced whose forecast held.

3.2 From quantity surveying and commercial

Transfers. Measurement, valuation, the contract, notices and time bars, and an instinct for entitlement. Genuine fluency in what may be claimed and what must be conceded. Familiarity with subcontract administration, which is where most cost risk actually lives.

Missing. Forward-looking forecasting. Quantity surveying is largely retrospective and periodic — what has been done, what is payable now. Controls is asked what the position will be at completion, on a stated assumption. Also missing, frequently: the logic-driven programme, as distinct from a bar chart of intent.

Learn first. Earned value mechanics and why earned value, certified value and recognised revenue are three different numbers; forecast method selection; critical-path logic well enough to interrogate a planner.

Characteristic failure. Treating the certified valuation as the performance measure, so the project looks exactly as good as the client's quantity surveyor is currently willing to admit.

Evidence of transfer. A forecast at completion you owned and defended; a reconciliation between the valuation position and the earned value position that explained the gap rather than closing it by assertion.

3.3 From finance and accounting

Transfers. The ledger, the control environment, accruals and cut-off — often the strongest cut-off discipline of any route. Segregation of duties, reconciliation as a reflex, and a healthy scepticism about numbers that arrive without a source. Comfort with cash as distinct from cost.

Missing. Physical progress. Financial systems record what has been transacted; controls must state what has been achieved, which no ledger knows. Also missing: schedule literacy, and the willingness to publish a number that is deliberately an estimate rather than a fact.

Learn first. Progress measurement and rules of credit; the earned value relationships and what they are and are not evidence of; how a programme is built and what makes a date credible.

Characteristic failure. Reading spend as progress, and reporting a project as under budget when it is simply late in placing its commitments.

Evidence of transfer. An earned value position you derived from raw quantities rather than received from the field; a variance narrative that identified a physical driver rather than a ledger movement.

3.4 From site supervision and construction management

Transfers. The most valuable and least teachable asset in the function: knowing what "70 % complete" looks like. Productivity intuition, crew and access logic, and the ability to tell a plausible progress claim from an implausible one. Credibility with the people who produce the data.

Missing. The formal apparatus — the data model, the systems, the arithmetic, the documentary discipline. Often, the habit of writing things down in a form someone else can audit.

Learn first. The WBS-to-CBS mapping and how the accounts work; earned value arithmetic; and the documentary habits of the function, particularly recording the basis of a judgement, not just the judgement.

Characteristic failure. Overriding the system with field knowledge without leaving a trace, so the report is right this month and unexplainable next month.

Evidence of transfer. A progress position you documented to rules of credit and defended against the subcontractor's claim; a forecast built from measured productivity rather than opinion.

3.5 From data and analytics

Transfers. Data modelling, reconciliation at volume, automation, and the ability to see errors that are invisible in a spreadsheet of the same data. Increasingly, the ability to work with the Institute's Domain 13 material on AI-assisted controls, where the constraint is rarely the model and almost always the data.

Missing. Domain meaning. A pipeline that joins cost to schedule perfectly can still produce nonsense if it joins a committed cost to an earned value at different data dates. Also missing: the contract, and the fact that some numbers are contested rather than merely uncertain.

Learn first. What each field actually means and when it is valid; the period cycle and its cut-off; the difference between a number that is wrong and a number that is disputed.

Characteristic failure. Building a dashboard that is faster, prettier and confidently incorrect, because nobody in the chain could tell them the accrual had not yet been posted.

Evidence of transfer. An automation that a controls manager relies on at month-end; a data-quality control you designed that caught a real error before it reached a report.

3.6 Directly from education

Transfers. Method, currency of technique, and no habits to unlearn. Graduates are frequently the only people in the room who have read the whole standard rather than the parts their employer uses.

Missing. Everything the other five routes carry: no quantity intuition, no ledger, no contract, no site. This is a real gap and pretending otherwise wastes everyone's time.

Learn first. One control account, end to end, every cycle, until you can do it without the template — then a second one in a different discipline. Depth on one account beats familiarity with all of them.

Characteristic failure. Producing formally correct numbers from inputs that are wrong, because there is no internal benchmark against which the input looks implausible.

Evidence of transfer. A cycle you ran unsupervised; an input you challenged and were right about.

4. What the interview is actually testing

Whatever route you took, two questions sit under the technical conversation.

Can you produce a defensible number under time pressure? Not a perfect number. Month-end does not wait, data is incomplete, and the skill is knowing which incompleteness matters. The answer that impresses is not "I would gather more data" but "I would report it with the accrual estimated on this basis, flag it as an estimate, and correct it next cycle".

Will you say an unwelcome number out loud? The function exists to tell an organisation something it may not want to hear. Every experienced interviewer is listening for evidence that you have done this and survived it. The Institute's position on the conduct of this is in `ETH-03`; here it is enough to say that an entrant who has never had an uncomfortable conversation about a forecast has not yet done the job.

5. Choosing a first specialism deliberately

Most people are allocated to cost or planning by whichever vacancy existed. It is worth choosing, because the two ladders differ in what they demand and what they defend.

Cost suits people who are comfortable with reconciliation, with incomplete data at cut-off, and with holding a position that will be tested against an audit trail. Planning suits people who think in dependencies and are prepared to have their work used as evidence in a dispute. Risk and estimating are usually entered from one of those two rather than directly. Change and commercial is the natural home for the quantity surveying route and the fastest route to understanding what the project is actually worth.

The pathways themselves are set out in `CAR-03 — Cost engineer to cost manager` and `CAR-04 — Planner to planning manager`. What matters at the point of entry is that you choose one to be genuinely good at, and stay conversationally fluent in the rest.

6. What the Institute's route offers, stated plainly

PCI certification is open to anyone with three years of professional experience, in any field. There is no degree requirement. Full-time-equivalent experience is counted, part-time counts pro rata, self-employment counts, and a career break does not reset the clock. That eligibility rule is deliberate: it fits a function that is entered sideways, and it means a route-changer is not excluded for having spent their first years elsewhere.

What it does not do is promise anything about employment. PCI is developed with reference to ISO/IEC 17024 personnel-certification principles and is not accredited. A credential evidences assessed knowledge against a published standard; the evidence that you can do the work is the portfolio, and that is your own to build.

7. How this goes wrong

Leading with the transferable skill and never retiring it. The route in is a bridge, not a position. An accountant who is still the person who reconciles, three cycles later, has not entered project controls; they have relocated their old job.

Underestimating the contract. Of all the gaps in §3, this is the one that produces the most expensive errors, because a controls position that ignores notice provisions and time bars can destroy entitlement that was otherwise recoverable. Every route except quantity surveying arrives short here.

Assuming the gap is knowledge when it is habit. Cut-off discipline, documenting the basis of a judgement and never altering a historical position are behaviours, not facts. They are learned by doing them wrongly once in front of an auditor.

Taking a role with no month-end. Some positions labelled project controls are reporting-assembly roles with no ownership of a control account. They teach the apparatus and none of the judgement, and they are difficult to leave because the evidence they generate is thin.

Waiting until you are competent to be useful. The fastest transfers happen when an incomer applies their existing strength to a real controls problem in the first cycle — the engineer who reworks the quantity basis, the analyst who finds the duplicated commitment — while learning the core underneath it.

8. Worked example — three routes, one control account, three conclusions

Illustrative figures. Currency-neutral units, one control account, at a single data date. Budget at completion 1,200,000. Invoiced cost 540,000. Goods and services received but not yet invoiced

60,000. Purchase orders raised to date 900,000. Physical progress, measured to agreed rules of credit, 55 %. Assumptions: the rules of credit are applied consistently; the 60,000 is the complete un-invoiced position at cut-off.

The finance-trained reading. Spend of 540,000 against a budget of 1,200,000 is 45 % of budget consumed. Conclusion: comfortably under budget.

The site-trained reading. The work is 55 % complete. Conclusion: ahead.

The controls reading. Actual cost = invoiced + accrued = 540,000 + 60,000 = **600,000**. Earned value = 55 % × 1,200,000 = **660,000**. Cost performance index (CPI) = earned value ÷ actual cost = 660,000 ÷ 600,000 = **1.10**. Forecast at completion on the assumption that performance persists = 1,200,000 ÷ 1.10 = **1,090,909**.

Then the question neither of the first two readings asks. Open commitment = purchase orders raised − cost incurred = 900,000 − 600,000 = **300,000**. Remaining budgeted scope = 45 % × 1,200,000 = **540,000**. So 540,000 − 300,000 = **240,000** of remaining scope has not yet been ordered — and the 1.10 index tells you nothing about what it will cost when it is, because none of it has been priced in the market yet.

All three readings are arithmetically correct. The finance reading is a true statement about cash. The site reading is a true statement about physical work. Only the third asks what the project will cost, and only the third notices that a fifth of the remaining scope is unpriced. That difference — the question asked, not the arithmetic used — is the whole of what a route-changer has to acquire.

9. Checklist — for anyone making the move

- Write down which of the six routes you are on, and read your own gap in §3 before your first interview.
- Learn the data model of your prospective employer first: WBS, CBS, code of accounts, and how they map to the finance system. Ask for it in week one; it is not confidential and nobody minds.
- Find out what the cut-off date is, what happens on it, and who has to have done what by then.
- Read the contract's change and notice clauses even if commercial "own" them. You will be asked to produce the number that supports a notice.
- Take ownership of one control account end to end, including the accrual, and do not delegate the part you find hardest.
- Each cycle, write your forecast down before the meeting and check it the following cycle. Keep the record; it becomes the evidence in `CAR-07`.
- Before your first month-end, plan the cycle with `CAR-08 — The first ninety days in a controls role`.

An incomer is judged on how quickly the tolerance for their gap becomes unnecessary. That is measured in cycles completed, not in months served, and it starts on the first cut-off you are responsible for.

Related

- `CAR-01 — The project controls career roadmap` — the map of the function this document feeds into
- `CAR-03 — Cost engineer to cost manager` — the cost ladder, if you choose that specialism
- `CAR-04 — Planner to planning manager` — the planning ladder, if you choose that one
- `CAR-07 — Building a portfolio of evidence` — how to capture the evidence named in §3
- `CAR-08 — The first ninety days in a controls role` — what to do once you are in
- `BPG-06 — Progress measurement and rules of credit` — the mechanics behind the worked example

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 1 (foundations of accounting for project controls), Domain 3 (budgeting and forecasting), Domain 5 (cost management and cost control), Domain 6 (earned value management and forecasting) and Domain 10 (project scheduling); and from the Institute's published eligibility rule of three years' professional experience in any field. No external market, salary, demand or hiring data has been used, and none should be inferred.

Status and version

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CAR-03 · CAREER ROADMAP

Cost engineer to cost manager

What is owned at each rung of the cost ladder, and the transition that blocks people

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

The cost ladder described by what is actually owned at each rung — the work, the artefacts, and the decisions a person is trusted to make — from cost analyst to cost manager. Its centre is the transition that stops most careers: moving from producing the report to owning the forecast, which means choosing the assumption a number rests on and being able to say what would change your mind.

About this document

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Contents

Cost engineer to cost manager

In one paragraph. This document sets out the cost side of a project controls career as a sequence of custodies: at each rung, the work performed, the artefacts owned, the decisions the holder is trusted to make, and what has to be demonstrated to leave. Its centre is the transition that stalls more cost careers than any other — the move from producing the monthly report to owning the forecast, which is not a change of arithmetic but a change of accountability, because a forecast is a chosen assumption with your name on it. No timescales are given for any rung; how long one takes depends on sector, employer and project size, and the useful question is what you can demonstrate.

Who this is for. Cost analysts, cost engineers, cost controllers, senior cost engineers and the project controls managers who develop them; also quantity surveyors and project accountants moving into cost control.

1. What the ladder actually measures

Not seniority, and not headcount. The cost ladder measures the size of the number you are trusted to be wrong about, and the degree to which that number is contested.

At the first rung, being wrong means a miscoded invoice that someone corrects. At the last, being wrong means an organisation made a capital decision on a forecast that did not hold. Everything between is a graduated increase in exposure, and every promotion is really a judgement by somebody that your exposure can safely grow.

That is why titles are unreliable. Cost engineer in one organisation owns a set of control accounts and their forecast; in another it means assembling a report from data produced elsewhere. Read the artefacts. The four questions that identify a rung, whatever the title on the door: which numbers do you produce, which do you sign, which do you defend, and which are produced by other people on a process you designed?

2. Rung one — cost analyst, cost assistant, junior cost engineer

The work. Coding and validating invoices to the code of accounts; allocating timesheets; maintaining the commitment register against raised purchase orders; assembling the report pack; keeping the cost ledger reconciled to the finance system; chasing the data that other people owe the cycle.

Artefacts owned. The commitment register. The cost-ledger-to-general-ledger reconciliation. The timesheet allocation. The data-quality log — the list of what was missing at cut-off and what was estimated in its place.

Decisions trusted. Which control account a transaction belongs to. Whether a reconciliation difference is a timing difference or an error. Whether the data is complete enough to publish, and if not, saying so before the report goes out rather than after.

What must be demonstrated to leave. That the numbers you assemble are complete, on time and traceable without supervision — and, critically, that you notice when they are not. The signal is unprompted: the person who arrives at the cost engineer's desk before cut-off to say "the site accommodation accrual looks low against last month, I think we are missing an invoice" is demonstrating the next rung while occupying this one.

The trap. This rung is where people learn the systems and mistake system fluency for cost control. A person who can operate every report in the tool and cannot say whether the position is complete has mastered the apparatus and none of the discipline.

3. Rung two — cost engineer

The work. Ownership of a defined set of control accounts. Each cycle: establish the accrual position, measure or verify progress, calculate earned value, compute the variance, and produce a forecast for the remaining scope with a stated basis. Between cycles: price change, maintain the trend register, and challenge commitments that do not match the scope.

Artefacts owned. The control-account forecast and its basis of estimate. The accrual schedule. The earned value calculation and the rules of credit applied to it. The variance narrative. The trend entries for their accounts.

Decisions trusted. Whether an accrual reflects work genuinely performed. Whether a variance is structural or a one-off — which is the decision that drives the forecast method. What the remaining scope will cost, and on what assumption. Whether a trend has become certain enough to move into the forecast.

What must be demonstrated to leave. A forecast, owned across several cycles, that either held or failed for a reason you predicted. The evidence is not accuracy alone; a forecast that was wrong for a named, unforeseeable reason is stronger evidence than one that was right by luck. See §6.

The trap. Producing the forecast the schedule implies, or the one the project manager expects, and calling it a forecast. This is the rung where the transition in §6 either happens or does not.

4. Rung three — senior cost engineer, cost lead

The work. Ownership of the total project cost position rather than a slice of it. Consolidating other people's control accounts and finding the inconsistencies between them. Choosing and defending the estimate at completion for the project. Owning the cash flow, the contingency position and the drawdown record. Presenting to the project director and to functional management, who will not have the same questions.

Artefacts owned. The project estimate at completion and its method rationale. The cash flow forecast. The contingency drawdown record. The consolidated trend register. The movement bridge between last period's forecast and this one.

Decisions trusted. Which forecasting method the project uses and why — the method-selection judgement that `BPG-09 — Estimate at completion` treats in depth. What is released from contingency and on whose authority. When a trend is escalated rather than absorbed. When to tell the project director that the number is going to move before it has moved.

What must be demonstrated to leave. That your position survives a hostile reading. Being right is necessary and no longer sufficient: at this rung the position is challenged by people with an incentive to prefer a different one, and the evidence is that you can decompose your own movement, name the driver of each line, and separate what is fact from what is judgement.

The trap. Defending the number instead of the reasoning. A senior cost engineer who cannot say which part of the forecast is a judgement, and therefore contestable, will eventually be either overridden or distrusted, and both are worse than conceding the judgement and holding the facts.

5. Rung four — cost manager

The work. Ownership of the cost function rather than the cost number. Designing the control environment: the cost breakdown structure and code of accounts, the estimating basis and its handover into control, the reporting calendar and cut-off, the change and trend thresholds, the delegated authorities, and the assurance regime that tests whether any of it is being followed. Building and developing the people. Interfacing with finance, procurement, commercial and the estimating group as a peer.

Artefacts owned. The cost control procedure. The code of accounts and its mapping to the enterprise resource planning system. The reporting calendar. The change and trend thresholds and the approval matrix. The assurance and health-check regime. The competency and succession position of the team.

Decisions trusted. What gets a budget and at what level of detail. What triggers a trend and what triggers an escalation. Who may move money between accounts, and up to what value. What the function will and will not report — including the decision to publish a number the project would rather delay. When to stop a cycle because the data is not fit to publish.

What must be demonstrated. That the function produces a defensible position when you are not in the room. The test is unglamorous and exact: take a month's leave and read the report that went out without you. If it is materially different in character from the one you would have produced, you have designed a role, not a function.

The trap. Continuing to be the best cost engineer on the project. It is the most understandable failure at this rung and the most damaging, because it guarantees the function's ceiling is your personal throughput.

6. The transition that actually blocks people

Between rung two and rung three sits a change most people are never told about: **from producing the report to owning the forecast.**

Producing means assembling correct arithmetic from inputs. It is honest, necessary work and it is completely learnable. Owning means four additional things, and every one of them is uncomfortable.

You choose the assumption. As `BPG-09` sets out, several forecasting methods are defensible on the same data and they give materially different answers. The arithmetic does not choose; you do. Owning the forecast means stating which behaviour of the remaining work you are assuming and why.

You can name what would change your mind. A forecast without a falsification condition is an opinion. "I am forecasting on the assumption that the productivity shortfall is access-driven and resolves when the third floor is released; if the rate has not recovered by the second week after release, this forecast moves up" is a position. The next sentence a competent director asks for is exactly that one.

You publish it before you are asked. This is the practical mechanism, and it is available to anyone at rung two without permission. Each cycle, write down your forecast for your accounts, with the assumption and the falsification condition, and date it. Next cycle, before you write the new one, check the old one and record why it moved. Within a handful of cycles you have something almost nobody else has: a documented record of your own forecasting judgement, including the times it was wrong and what you learned. That record is the argument for the next rung, and it is portable evidence in the sense of `CAR-07`.

You carry it into the room. A forecast that lives in a file is not owned. Owning includes presenting it to someone who would prefer a lower number, and not adjusting it in the meeting. Where that pressure crosses into a request to misstate, it stops being a career question and becomes a conduct one; the Institute's treatment is in `ETH-03 — Reporting numbers under pressure`.

The reason this transition blocks people is structural: nobody hands over forecast ownership, because on paper it already belongs to someone senior. It is taken by doing it in parallel until the formal handover is a formality.

7. How this goes wrong

Forecasting the budget. The most common failure in the function: a forecast that equals the budget for every account that has not yet visibly failed. It reports nothing, absorbs no learning, and collapses in one step near the end when the truth becomes undeniable.

Letting the accrual float. An incomplete accrual makes cost performance look better and then worse, and the swing is mistaken for a real event. `BPG-07 — Accruals and cut-off discipline` owns this topic; its career relevance is that accrual discipline is the single fastest way for a cost engineer to become trusted, because it is visible, checkable and frequently neglected.

Confusing contingency with headroom. Drawing contingency to cover an ordinary overrun, without recording why, destroys the only record of what the contingency was for. The drawdown record is treated in `BPG-10 — Contingency and management reserve`; the career point is that a cost lead who cannot explain where contingency went has forfeited the argument about how much is needed.

Hiding the movement. Restating a prior period so this period looks stable is the fastest way to lose the room permanently. The correct instrument is a movement bridge that shows exactly what changed and why — see `CAR-05` §6.

Becoming a reporting function. Some cost teams spend the entire cycle producing the report and none of it analysing the result. The symptom is a report that is beautifully formatted and contains no statement about what will happen next.

Growing the team instead of the control environment. At rung four, adding people to absorb a data problem hides the problem and doubles its cost. The alternative is to fix the mapping, the cut-off or the rules of credit that made the data unreliable in the first place.

8. Worked example — the difference between producing and owning

Illustrative figures. Currency-neutral units, one control account at a single data date, rounded to whole units. Budget at completion (BAC) 4,000,000. Earned value (EV) 1,600,000. Actual cost (AC) 1,840,000. Planned value (PV) 1,800,000. Assumptions: the accrual is complete at the data date; progress is measured to agreed rules of credit; no approved change is pending.

The indices. Cost performance index: CPI = EV ÷ AC = 1,600,000 ÷ 1,840,000 = **0.870** (3 decimal places). Schedule performance index: SPI = EV ÷ PV = 1,600,000 ÷ 1,800,000 = **0.889**.

Three defensible forecasts on the same data.

Performance persists: EAC = BAC ÷ CPI = 4,000,000 ÷ (1,600,000 ÷ 1,840,000) = 4,000,000 × 1,840,000 ÷ 1,600,000 = **4,600,000**. (The unrounded index is used throughout; substituting the rounded 0.870 would give 4,597,701, a difference of about 2,300 and a useful reminder that indices are rounded for reporting, never for calculation.)

The overrun was a one-off, remaining work performs to budget: EAC = AC + (BAC − EV) = 1,840,000 + (4,000,000 − 1,600,000) = 1,840,000 + 2,400,000 = **4,240,000**.

Cost and schedule pressure both persist: EAC = AC + (BAC − EV) ÷ (CPI × SPI). The denominator is 0.86957 × 0.88889 = 0.77295, so (BAC − EV) ÷ 0.77295 = 2,400,000 ÷ 0.77295 = 3,105,000, and EAC = 1,840,000 + 3,105,000 = **4,945,000**.

The spread. 4,945,000 − 4,240,000 = **705,000** on identical inputs — about 17.6 % of the budget (705,000 ÷ 4,000,000 = 0.176). Method selection is treated in `BPG-09`; what matters here is what the spread means for a career.

Producing is arriving at the meeting with the three numbers and asking which one to use. It is correct, it is useless, and it transfers the decision upwards.

Owning is arriving with one number and four sentences: the forecast is 4,945,000; the overrun is structural because the productivity shortfall on the two largest work packages has persisted across three consecutive cycles and is present in both the timesheet and the quantity data; if the shortfall is access-driven it should recover within two cycles of the area handover, and I will report the achieved rate against 0.87 each cycle; if the rate has not moved by the second cycle after handover I will move the forecast again and I will tell you before the report goes out.

Same arithmetic. Entirely different job. The second version can be wrong and still be the reason someone is promoted, because it is checkable — and the person who published a falsification condition and honoured it has demonstrated the one thing the next rung requires.

9. Checklist — evidence that you are at the next rung already

- You produce a written forecast for your accounts each cycle, with an assumption and a falsification condition, before anyone asks for it.
- You can explain last cycle's movement in causes, not in categories, and you keep the record.
- Your accrual position has survived at least one audit or one challenge from finance.
- You have identified a variance driver from the physical data before it appeared in the cost ledger.

- You have told a project manager a number they did not want, in advance, with the evidence attached.
- You can state which part of your forecast is fact and which is judgement, and you volunteer the distinction rather than being asked for it.
- You have designed or repaired one control — a mapping, a cut-off rule, a rule of credit — that stopped a class of error rather than one instance of it.
- You have a dated record of a forecast you got wrong, and a written account of why.

The last item is the one people leave out, and it is the one that persuades. A cost professional with no record of having been wrong has either not been forecasting or has not been checking, and an experienced interviewer can tell which within two questions.

Related

- `CAR-01 – The project controls career roadmap` — where this ladder sits in the whole function
- `CAR-04 – Planner to planning manager` — the parallel ladder, and its own blocking transition
- `CAR-05 – Into project controls leadership` — what rung four leads to
- `CAR-07 – Building a portfolio of evidence` — how to capture the forecast record described in §6
- `BPG-07 – Accruals and cut-off discipline` — the mechanics behind the accrual judgements at rung two
- `BPG-09 – Estimate at completion – choosing and defending a method` — the method selection referenced in §8
- `BPG-10 – Contingency and management reserve` — contingency governance at rungs three and four

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 3 (budgeting and forecasting), Domain 4 (performance management, variance analysis and management reporting), Domain 5 (cost management and cost control), Domain 6 (earned value management and forecasting) and Domain 7 (contracts, commercial management, invoicing and revenue); and from the PCL-AI competency set. Earned value relationships are described in the Institute's own words and no protected text is reproduced. No pay, market or demand data has been used — for the Institute's position on market data see `SAL-01` and `SAL-06`.

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CAR-04 · CAREER ROADMAP

Planner to planning manager

What is owned at each rung of the planning ladder, and why defending a programme is a different job from maintaining one

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

The planning and scheduling ladder described by what each rung owns — the work, the artefacts, the decisions trusted, and what must be demonstrated to leave. Its centre is the transition that stalls most planning careers: moving from maintaining the programme to defending it, which means being able to explain every movement between updates in causes rather than in dates, and keeping the record that makes the explanation checkable.

About this document

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Contents

Planner to planning manager

In one paragraph. A programme is used for three different purposes — to sequence work, to measure progress, and to prove entitlement — and most planners are trained for the first while ultimately being judged on the third. This document sets out the planning ladder as a sequence of custodies: at each rung, the work, the artefacts owned, the decisions trusted, and what must be demonstrated to leave. Its centre is the transition that stalls planning careers, from maintaining the programme to defending it, and it gives the specific habit that makes the transition visible: a written delta narrative, every update, explaining movement in causes rather than dates.

Who this is for. Schedulers, planners, planning engineers, lead planners and planning managers, and the project controls managers who develop them; also cost professionals who need to interrogate a programme rather than accept it.

1. Three jobs the same programme has to do

The programme is the only artefact in project controls that is simultaneously an operational instrument, a measurement baseline and a piece of evidence.

To sequence work. The delivery team needs to know what happens next, in what order, with what resources. This use rewards detail, currency and readability.

To measure progress. Earned value, the s-curve and the forecast all depend on the programme's activity structure and its link to the cost breakdown. This use rewards stability, consistent coding and disciplined data dates.

To prove entitlement. When a project is delayed, the programme and its update history become the factual record from which cause and effect are argued. This use rewards logic that reflects real constraint, updates that were not retrospectively edited, and a narrative written at the time.

These three uses conflict. Detail helps the first and harms the second. Re-sequencing to help the delivery team can destroy the evidence needed by the third. Recognising and managing the conflict is what distinguishes the upper rungs of this ladder; the failure to recognise it is what produces a programme that is popular on site and useless in a dispute.

2. Rung one — scheduler, assistant planner

The work. Collecting actual start and finish dates; recording progress against activities; maintaining the short-term lookahead; producing the s-curve and the standard reporting outputs; keeping the activity coding correct; distributing the update.

Artefacts owned. The progress update file. The lookahead. The s-curve and the standard report outputs. The record of what was reported by whom.

Decisions trusted. Whether a reported percentage is credible against what the lookahead said would happen. Whether an activity is genuinely started or merely mobilised. Whether the update is complete enough to run.

What must be demonstrated to leave. That the update is right, on time and reproducible — and that you challenge implausible progress rather than entering it. The signal is a scheduler who returns to a supervisor with "you reported 60 % last period and 62 % now, but the lookahead had this activity finishing; what changed?" That question is the beginning of planning.

The trap. Becoming the person who operates the software. Tool fluency is assumed everywhere above this rung and distinguishes nobody.

3. Rung two — planning engineer

The work. Ownership of a section of the programme: building the logic, agreeing durations with the people who will perform the work, mapping activities to the work breakdown structure and to cost, running the update including the decisions the software cannot make, and writing the variance-to-baseline narrative for that section.

Artefacts owned. The section's logic and durations, with the basis on which each was agreed. The update package. The schedule health metrics for the section — open ends, constraints, negative lag, out-of-sequence progress, excessive durations (`BPG-05 — Schedule quality` owns the metric set). The section's variance narrative.

Decisions trusted. Whether to accept a subcontractor's declared duration or challenge it, and on what evidence. Whether progress that occurred out of sequence should be handled by retained logic or progress override, which changes the answer the programme gives. Whether an apparent recovery is real or an artefact of how the update was run. Whether to re-sequence or to report the slippage.

What must be demonstrated to leave. That you can explain the section's movement between two updates in causes, and that the explanation is supported by the record. This is the beginning of the transition in §6.

The trap. Accepting durations because they were given by someone senior or by a subcontractor with an interest in them. A duration accepted without a basis becomes, three updates later, a date you cannot defend and did not choose.

4. Rung three — senior planner, lead planner

The work. Ownership of the integrated programme: all sections, all interfaces, all contributors, including the subcontractor programmes that must be accepted or rejected. Ownership of the baseline and its change control. Running quantitative schedule risk analysis (QSRA) and interpreting it. Choosing the delay analysis approach when time becomes contested, and working with the commercial team on notices and entitlement.

Artefacts owned. The integrated baseline and the baseline change log. The driving-path narrative each period. The QSRA model, its input basis and its outputs (`BPG-17` owns the technique). The schedule specification applied to subcontractor programmes. The evidence set: dated, unaltered update files.

Decisions trusted. What the critical path actually is when the software's answer is an artefact of a constraint or a calendar. When a rebaseline is legitimate and when it would erase the evidence of delay — one of the most consequential judgements in the function. What float is real and who owns it under the contract. When to escalate a date movement rather than absorb it in the lookahead.

What must be demonstrated to leave. That the programme has survived challenge — from a client, an auditor, a subcontractor or a claim — and that the record supported it. This is the rung where planning becomes a legal-adjacent discipline, and the evidence is documentary.

The trap. Logic churn. Changing links between updates to make the answer look sensible destroys comparability, and comparability is the whole basis of a delay argument. If the logic must change, it changes as a recorded, reasoned amendment, not silently.

5. Rung four — planning manager

The work. Ownership of the planning function: the schedule specification and level-of-detail standard, the calendars and coding structures, the update procedure and its cut-off, the tooling, the assurance regime that tests compliance, and the contractual position of the programme itself. Developing planners. Negotiating what the programme is contractually required to be, and resisting requirements that would make it unusable.

Artefacts owned. The schedule specification and planning procedure. The coding and calendar standards. The update calendar and its cut-off. The assurance and health-check regime. The subcontractor programme acceptance criteria. The function's competency and succession position.

Decisions trusted. The level of detail the programme will be built to — the decision that determines whether it can serve all three purposes in §1. Who may change logic, and under what record. What is submitted contractually and what is retained as internal working. Whether to accept a client-imposed schedule requirement, and what to say when it is unworkable.

What must be demonstrated. That an update produced in your absence is defensible. Same test as the cost ladder in `CAR-03` §5, and the same failure if you remain the best planner on the project rather than the person who made good planning reproducible.

6. The transition that actually blocks people

Between rung two and rung three sits the change nobody names: **from maintaining the programme to defending it.**

Maintaining is running the cycle correctly — statusing, recalculating, distributing. Defending is being able to answer, from the record, why the programme says what it says. Four capabilities separate them.

Every driving link has a reason you can state. Not "that is how it was built". A defensible logic tie reflects a physical, contractual or resource constraint that someone can be named for. Planners who inherit a programme and never interrogate its logic are defending someone else's assertions.

Movement is explained in causes, not in dates. "Completion has moved out by three weeks" is a measurement. "Completion has moved out by fifteen working days because containment installation is achieving 22 m per working day against a planned 30, and the remaining quantity at the achieved rate takes fifteen days longer than the baseline allowed" is a defence — and §9 works that case through. The instrument is a written delta narrative attached to every update: what moved, by how much, driven by what, and what would reverse it.

The record is unaltered and dated. Every update file retained as issued, never overwritten; the narrative written at the time, not reconstructed. The value of a contemporaneous record is that it was written before anyone knew which way the dispute would go, and that value is destroyed by a single retrospective edit.

You separate the measurement from the promise. A programme that shows recovery contains a promise someone must keep. If you report recovery because a delivery manager asserted it, you have made their promise on your document. Defending means the recovery is either evidenced by an achieved rate or labelled as an assumption with a named owner and a date by which it must be visible.

The reason this transition blocks people is that maintaining is measurable and defending is not — until the moment it is tested, at which point it is the only thing that matters. As with the cost ladder, the route through is to do it before you are asked: write the delta narrative every update, whether or not anyone reads it, and keep it.

7. Working with cost, and why it doubles your value

The programme and the cost position describe the same project and are usually produced by different people using different structures. Two things follow.

A planner who understands earned value can tell whether a schedule movement should have a cost consequence and whether the reported cost position reflects it. A planner who cannot will be surprised when the cost team forecasts an overrun that the programme said nothing about.

More practically, the integration point is the control account. If the activity structure does not map cleanly to the cost breakdown, every reconciliation between programme and cost becomes manual, opinionated and slow. That mapping is a rung-three and rung-four decision, and it is where planners have the largest effect on the quality of everything else the function produces.

8. How this goes wrong

Detail as a substitute for logic. A programme with tens of thousands of activities and no defensible driving path is harder to maintain and no more credible. Level of detail should be set by what must be measured and controlled, not by what can be listed.

Constraints used to hold dates. Applying a constraint to make an activity report the required date suppresses the very signal the programme exists to give. The date is met on paper and the delay is discovered when it is too late to mitigate.

Rebaselining to make the variance disappear. A rebaseline is legitimate when scope or an approved change makes the old baseline meaningless. It is illegitimate when its purpose is to reset the variance, and it is close to fatal when time is contested, because it removes the reference against which delay would have been measured.

Progress override applied without thought. Out-of-sequence progress handled by the wrong setting can transform the critical path and the completion date without anyone deciding anything. This is a judgement, made every update, and it must be recorded.

Accepting the subcontractor's programme unexamined. An unexamined subcontract programme imports its optimism and its missing interfaces into your integrated model, and you inherit both.

Writing the narrative after the outcome is known. Reconstructed narratives are recognisable and they destroy credibility precisely when it is needed. Write it at the update, before you know what happens next.

9. Worked example — defending a movement between two updates

Illustrative figures. One driving activity, cable containment installation, on a project with a five-day working week. Assumptions: the installation is the sole driver of the completion date; the planned rate is the rate used to build the baseline duration; no scope change occurred in the period; the period between update 7 and update 8 is 20 working days.

Position at update 7. Remaining quantity 1,200 m. Planned rate 30 m per working day. Remaining duration = 1,200 ÷ 30 = **40 working days**.

What the plan implied for the period. 20 working days × 30 m = **600 m** installed, leaving 600 m and 20 working days remaining at update 8.

What actually happened. 440 m installed in 20 working days. Achieved rate = 440 ÷ 20 = **22 m per working day**. Remaining quantity = 1,200 − 440 = **760 m**.

Two defensible remaining durations, and they differ by nine days. Assuming the planned rate is recovered: 760 ÷ 30 = 25.3 → **26 working days**, which is 6 days more than the 20 the plan implied. Assuming the achieved rate persists: 760 ÷ 22 = 34.5 → **35 working days**, which is 15 days more than planned. Difference between the two positions = 35 − 26 = **9 working days**.

The plausibility test that turns a choice into a defence. To finish in the 20 working days the baseline still shows as remaining, the crew would have to achieve 760 ÷ 20 = **38 m per working day** — that is 38 ÷ 30 = 1.267, or **26.7 % above the planned rate**, and 38 ÷ 22 = 1.727, or **72.7 % above the rate actually being achieved**. Nobody has yet demonstrated the planned rate, let alone a quarter more than it.

The defence. Report 35 working days remaining and 15 days of movement; state the mechanism — the achieved rate is 22 against a planned 30, evidenced in both the installed quantity record and the timesheets; state the recovery condition — a second crew, or release of the two areas currently permit restricted, would raise the achievable rate, and the required rate to recover fully is 38 m per day, which is above anything demonstrated on this project; and state the test — the achieved rate will be reported against 30 every update, and if it has not exceeded 30 within two updates of the areas being released, the recovery assumption is withdrawn.

If instead you report 26 days because a delivery manager has committed to recovery, you have not made a forecast. You have signed someone else's promise, on your document, and it is your credibility that is spent when it is not kept.

10. Checklist — evidence that you are at the next rung already

- You attach a written delta narrative to every update, explaining movement in causes, and you keep the narratives.
- You can name the physical, contractual or resource reason for every link on the current driving path.
- You retain every issued update file unaltered, and you can produce the set for any period on request.
- You have rejected or challenged a declared duration or a reported percentage and been right.
- You have run the plausibility test in §9 on a recovery claim, in writing, before agreeing to show it.
- You have recorded an out-of-sequence judgement — retained logic or progress override — and its effect on the completion date.
- You can explain the mapping between your activity structure and the cost breakdown structure, and you know where it breaks.
- You have told a project director that a date has moved before they heard it from anyone else.

A planner whose updates are correct but whose movements are unexplained is discovered in the first contested month, usually by someone external, and by then the record that would have defended them was never written.

Related

- `CAR-01 — The project controls career roadmap` — where this ladder sits in the whole function
- `CAR-03 — Cost engineer to cost manager` — the parallel ladder and its own blocking transition
- `CAR-05 — Into project controls leadership` — what rung four leads to
- `CAR-07 — Building a portfolio of evidence` — how to capture the update record described in §6
- `BPG-05 — Schedule quality: a practical review` — the metric set referenced at rung two
- `BPG-12 — Claims and extension of time` — where the evidential use of the programme is treated in full
- `BPG-17 — Quantitative schedule risk analysis` — the technique owned at rung three

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 6 (earned value management and forecasting), Domain 8 (project management lifecycle), Domain 10 (project scheduling) and Domain 12 (risk management for project controls); and from the PCL-AI competency set. Scheduling and delay concepts are described in the Institute's own words; no protected standard text, table or diagram is reproduced. No pay, market or demand data has been used.

Status and version

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CAR-05 · CAREER ROADMAP

Into project controls leadership

The move from being right to being believed, and from owning a number to owning the system that produces it

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Below leadership, a controls professional is measured on the accuracy of what they produce. At leadership they are measured on whether the organisation acts on it. This document sets out what changes: the four accountabilities of a controls leader, what a controls strategy actually decides, how a function is run rather than performed, how the relationship with the project director is constructed in advance, and the specific instrument — the movement bridge — that converts accuracy into credibility.

About this document

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Unresolved placeholders	0

Contents

Into project controls leadership

In one paragraph. The move into project controls leadership is not a larger version of the job below it. Below leadership you are measured on the accuracy of what you produce; at leadership you are measured on whether the organisation acts on it in time, and on whether the function can produce a defensible position without you. This document sets out the four accountabilities that come with the role, what a controls strategy actually decides, how to run a function rather than perform one, how to construct the relationship with the project director before you need it, and the single instrument — a movement bridge — that most reliably converts being right into being believed.

Who this is for. Cost managers, planning managers, project controls managers and heads of function, and the senior cost engineers and lead planners preparing for that step; also project directors who want to know what to expect from the role.

1. What changes

At rung three of either technical ladder — `CAR-03` for cost, `CAR-04` for planning — the job is to hold a defensible position. At leadership, three things change at once.

Your output is a decision, not a report. A correct forecast that arrives after the procurement commitment has been made has failed, however accurate it was. Timeliness is now part of correctness, and the calendar is a design decision you own.

You are accountable for numbers you did not check. The position now aggregates work by people you hired, using a process you designed. You cannot verify it all, so you must design controls that make undetected error unlikely and detected error early.

Being right becomes the entry ticket. It is assumed. What is now scarce is the ability to make an organisation act on an unwelcome number, which is a function of credibility, timing and the form in which the number arrives — not of its accuracy.

The failure of many strong technical practitioners at this step is to double down on the thing that got them here. They produce better analysis for a room that has already stopped listening, and conclude that the organisation is unserious. Occasionally that is true. More often the analysis was correct, late, unbridged and delivered to someone who had not been prepared for it.

2. The four accountabilities

That the position is true. Complete, reconciled, on a stated data date, with judgements labelled as judgements. This is inherited from the technical ladder and is non-negotiable.

That it arrives in time to change something. The reporting calendar is designed backwards from the decisions it must inform — the procurement commitment, the board review, the client submission — not forwards from when the data happens to be available. If the cycle cannot serve the decision, the cycle is wrong.

That the function can produce it without you. Documented procedure, trained people, a deputy who has actually run a cycle, and a review discipline in which someone other than the author checks the number. The measure is the report that goes out while you are on leave.

That the people in it get better. Every cycle is a training opportunity that is usually wasted. The mechanism is giving people numbers to own, early, with a review — and then defending their number in the room rather than replacing it, which is how ownership becomes real.

3. Setting a controls strategy

A controls strategy is not a statement of intent. It is a set of decisions that, once made, determine what the function can and cannot tell anyone. The instrument is a controls execution plan — `TPL-01` provides the template and `BPG-01 — Building a project controls function from zero` covers standing one up — and these are the decisions inside it that a leader personally owns.

The data model. The work breakdown structure, the cost breakdown structure and the code of accounts, and the mapping between them, the contract's payment structure and the finance system. Get this wrong and every reconciliation for the life of the project is manual.

Level of detail. Set by what must be separately measured and controlled. Too coarse and variance cannot be attributed; too fine and the cycle cannot be completed in time to matter.

Progress rules. Which rules of credit apply to which types of work, agreed before anyone claims anything. Retrofitting rules of credit to work already claimed is an argument nobody wins.

The calendar and the cut-off. Data date, cut-off time, who owes what by when, when the report publishes and into which decision. Published in advance and enforced.

Change and trend thresholds. What value triggers a trend, what triggers a change order, who approves at each level, and what may never be absorbed silently.

Contingency governance. Who holds it, who may release it, on what evidence, and how the drawdown is recorded.

Reporting hierarchy. What the project director sees weekly, what the board sees monthly, what the client sees contractually — and the rule that these are views of one position, never three positions.

The assurance regime. What is checked, by whom, how often, and what happens when a check fails.

Two of these are frequently deferred and should never be: the data model and the progress rules. Both become effectively unchangeable once work is under way.

4. Running a function rather than performing one

Capacity against cycle demand. A controls function has a sharply peaked workload. Plan the peak, not the average, and know which activities can be moved out of the peak — commitment reconciliation and trend review can; accrual cannot.

Deputising properly. A deputy who has never run the cycle is not a deputy. Someone else runs a full cycle, with you available and not intervening, before you need them to.

The review discipline. Someone other than the author checks every published number. This is not distrust; it is the same principle as any control environment. In practice it catches mapping errors, data-date mismatches and stale accruals — the errors that are invisible to the person who made them.

Hiring for the gap in the map. `CAR-01` §2 sets out the six disciplines. Hire against the discipline your function is thin in, and against the entry route (`CAR-02`) whose strengths you lack — not against the title of the person who left.

Developing by handing over exposure. The transition described in `CAR-03` §6 and `CAR-04` §6 is something you can grant. Give a cost engineer the forecast, or a planning engineer the driving-path narrative, before they are obviously ready, and review it rather than rewriting it.

5. The relationship with the project director

This relationship is constructed, not discovered, and the construction happens before the first difficult number, not during it.

Agree the escalation contract in the first weeks. What you will bring immediately, what waits for the report, and what you will never sit on. A workable default: any movement above an agreed threshold, any event that changes the completion date, and any number that has become uncertain enough that you would not sign it — all reported the day you know, not at month-end.

Never surprise them in public. A director who hears a bad number first in front of their own leadership will remember it far longer than the number. The corollary is the rule that no new figure appears in a meeting for the first time: if it is material, it was pre-briefed.

Bring the number with the decisions attached. A forecast movement presented alone invites the question "what are you doing about it", which is not your decision to make alone. A movement presented with two or three options, their cost, and the date by which each stops being available, is a management input.

Separate the number from the recommendation, explicitly. "The forecast is X" and "I recommend Y" are different statements with different authority. Conflating them lets a disagreement about the recommendation contaminate the number, which is the beginning of the pressure described in `ETH-03 — Reporting numbers under pressure`.

Know where the line is. Being asked to present a number differently, to defer publication by a week, or to show a sensitivity is ordinary management. Being asked to state a position you believe to be untrue is not, and it is a conduct matter with a defined route. A controls leader who has not thought about that line in advance will meet it unprepared.

6. Being believed: the movement bridge and the habits around it

Credibility in this function is built from a small number of unglamorous behaviours.

Reconcile to one source. Cost, commercial, schedule and risk report from the same data at the same data date. Where they differ, the difference is explained in the report, not resolved privately.

Never move a number silently. Every movement is bridged. The bridge — worked through in §8 — decomposes the change between last period's position and this one into named causes, and separates fact from judgement. Its effect on a meeting is out of proportion to the effort: it converts "why has your number gone up again" into "do we accept this one assumption", which is a discussion you can have.

Pre-brief the people who will be asked. Anyone who will be questioned about the number in a forum you are not in should have heard it from you first, with the same explanation.

Be first to report your own error. Nothing purchases credibility as cheaply, and nothing destroys it as thoroughly as an error found by someone else that you already knew about. The correct form is the error, its cause, its effect on the reported position, and the control that will stop it recurring.

Say the same thing to everyone. Different audiences get different levels of detail, never different positions. The moment two versions exist, the function's independence is gone and cannot be rebuilt on that project.

When you are overruled, record it and move on. Note the position you gave, the decision taken and the date, in the ordinary record rather than in a private file. Do not re-litigate it in every subsequent meeting, and do not leak it. If the decision requires you to publish something untrue, that is the different matter described in §5.

7. How this goes wrong

Remaining the best analyst in the room. The most common failure. The function's output ceiling becomes your personal throughput, the team never develops, and the position collapses the moment you are unavailable.

Building a report pack nobody reads. Volume is mistaken for rigour. `BPG-14 — Monthly reporting that gets read` owns the remedy; the leadership point is that a report is an instrument of decision, and if no decision has ever changed because of it, its length is not the problem.

Reorganising before diagnosing. New controls leaders frequently restructure the team in the first cycle. Almost always the problem is the data model, the cut-off or the rules of credit, and a reorganisation moves the same broken process between different people.

Confusing independence with distance. A controls function that is never on site, never in the procurement meeting and never in the subcontract negotiation is not independent; it is uninformed, and its numbers arrive too late to alter anything.

Managing the message instead of the position. Softening a number to preserve a relationship works exactly once. The second time, the audience discounts everything the function says, including the numbers that were never softened.

Never developing a successor. A leader who cannot be promoted because nobody can replace them has optimised for indispensability, which is a ceiling disguised as security.

Assuming the credential does the persuading. The Institute certifies knowledge against a published standard; PCI is developed with reference to ISO/IEC 17024 personnel-certification principles and is not accredited. Credibility in a project meeting is built from bridged numbers and kept promises, and nothing else substitutes.

8. Worked example — the movement bridge

Illustrative figures. Currency-neutral units, one project, month-on-month, rounded to the nearest thousand. Assumptions: same data date convention both periods; no change to the contract price;

the risk provision and the forecast are maintained separately so that a released risk reduces the forecast rather than the budget.

Last period's forecast final cost: 24,600,000. This period's: 25,380,000. Movement: +780,000.

Presented like that, the meeting is about 780,000 and about you. Bridged, it is about one number.

Line	Cause	Movement	Fact or judgement
1	Variations approved in the period, priced and instructed	+310,000	Fact
2	Scope identified as omitted from the estimate at handover, now quantified	+120,000	Fact
3	Productivity deterioration on two control accounts, forecast at the achieved rate	+295,000	Judgement
4	Risk retired without occurring; provision released from the forecast	−90,000	Fact
5	Subcontract re-let above the estimate carried in the forecast, award executed	+145,000	Fact
	Total movement	**+780,000**	

Check the bridge ties. 310,000 + 120,000 = 430,000; 430,000 + 295,000 = 725,000; 725,000 − 90,000 = 635,000; 635,000 + 145,000 = **780,000**. And 24,600,000 + 780,000 = **25,380,000**, which is the reported position. A bridge that does not tie exactly to the reported movement is worse than no bridge.

What the bridge does to the conversation. Four of the five lines are facts — instructions issued, scope quantified, a risk that did not occur, a subcontract awarded at a known value. They are not arguable, only regrettable. One line is a forecasting judgement: 295,000 ÷ 780,000 = **37.8 %** of the movement. The other 62.2 % is arithmetic on things that have already happened.

So the meeting should be about 295,000, and specifically about one question: is the productivity shortfall structural or recoverable? That question has an owner, an evidence base and a test date, and it can be answered. "Why has your number gone up by 780,000" cannot be answered at all, which is why the unbridged version of this conversation is where credibility is lost.

The second-order use. Keep the bridges. Twelve of them are a record of how your forecast behaved across a project: which lines recurred, which judgements were vindicated, how much of the total movement was ever within the function's control. That record is a leadership evidence set in the sense of `CAR-07`, and there are very few better answers to "how do you know your forecasting is any good".

9. Checklist — before you take the role, and in the first cycle after

- Read the contract yourself, including the payment mechanism, change mechanism and notice provisions. Do not accept a summary.
- Establish the escalation contract with the project director in writing, however informally, in the first weeks.

- Confirm the data model and the progress rules. If either is wrong, that is the first thing you change and the only thing worth restructuring for.
- Publish the reporting calendar backwards from the decisions it must serve, and enforce the cut-off once.
- Name your deputy and schedule the cycle they will run without you.
- Institute the review rule: no published number unchecked by someone other than its author.
- Start the movement bridge in your first reporting cycle, even if the movement is small. It is far harder to introduce during a difficult month.
- Identify the one control that would have caught the last unpleasant surprise on this project, and install it.
- Decide, before you are tested, what you will do if you are asked to publish a position you believe to be untrue.

A controls leader is ultimately judged on a single counterfactual: did the organisation make a different and better decision because the function existed? Accuracy is how you earn the right to be asked. Being believed is how the answer becomes yes.

Related

- `CAR-03 — Cost engineer to cost manager` — the cost ladder that leads here
- `CAR-04 — Planner to planning manager` — the planning ladder that leads here
- `CAR-07 — Building a portfolio of evidence` — how to evidence leadership work, including the bridges
- `CAR-08 — The first ninety days in a controls role` — the entry plan, which applies with more force at this level
- `BPG-01 — Building a project controls function from zero` — the standing-up problem in full
- `BPG-14 — Monthly reporting that gets read` — the reporting craft assumed in §7
- `TPL-01 — Project controls execution plan` — the instrument that records the decisions in §3

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 4 (performance management, variance analysis and management reporting), Domain 8 (project management lifecycle) and Domain 12 (risk management for project controls); and from the PCL-AI competency set, in particular project controls governance, performance measurement and human validation. The Institute's accreditation status is stated as published: developed with reference to ISO/IEC 17024 personnel-certification principles, not accredited. No pay, market or demand data has been used; see `SAL-01` and `SAL-06`.

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CAR-06 · CAREER ROADMAP

Moving between sectors

What genuinely transfers between EPC, oil and gas, rail, data centre, power and building — and what has to be relearned

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

Summary

Project controls method transfers between sectors almost completely; the intuition that makes a practitioner fast does not. This document gives five variables that characterise any sector — contract form, unit of production, cost shape, assurance overlay and decision cadence — then works through six sectors on those variables, naming what transfers, what must be relearned, and the mistake that identifies an incomer. It ends with the arithmetic of a benchmark that was correct in one sector and wrong in the next.

About this document

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Contents

Moving between sectors

In one paragraph. Cut-off discipline, earned value mechanics, forecast defence and change control work identically in every sector; what does not transfer is the intuition about magnitudes that makes an experienced practitioner fast, and that intuition is the first thing to distrust after a move. This document characterises a sector by five variables — the contract form and who carries risk, the unit of production, the shape of the cost, the assurance overlay, and the decision cadence — then applies them to six sectors, naming what transfers, what must be relearned, and the characteristic error of an incomer. It contains no claims about demand, hiring or pay in any sector.

Who this is for. Cost engineers, planners, risk analysts and controls managers considering or making a sector move, and the hiring managers deciding how much sector-specific knowledge genuinely matters.

1. Five variables that characterise any sector

Sectors feel very different from the inside and are structurally similar from the outside. If you can answer these five questions about a sector, you can work in it; if you cannot, no amount of general controls skill will stop you being surprised.

What is the contract form, and who carries which risk? Lump sum, remeasurable, reimbursable, target cost with a pain-and-gain mechanism, or a package structure combining several. This determines what change is, what may be claimed, what evidence is required, and therefore what the controls function spends its time doing.

What is the unit of production, and how is progress credited? Metres of pipe, tonnes of steel, engineering deliverables, procurement milestones, commissioning systems, or completed measured items in a bill of quantities. This determines the rules of credit and where progress can be inflated.

What shape is the cost? Labour-dominated, equipment-dominated, materials-dominated, or dominated by plant supply and long-lead items. A labour-dominated project is controlled through productivity; an equipment-dominated one is controlled through procurement and delivery dates, and the two demand different weekly attention.

What is the assurance overlay, and who audits you? Internal audit only, a client's assurance team, a regulator, a public-funding stage-gate regime, or a lender's technical adviser. This determines how much of your work is documentation of the work.

What is the decision cadence? Monthly reporting into a steering group, weekly into a delivery board, or hourly during a fixed-duration outage. Cadence determines the design of the whole controls cycle, and it is the variable incomers most often fail to adjust.

2. What transfers, and the one thing that does not

Transfers essentially unchanged: the data model and the control account as the unit of everything; the period cycle and cut-off discipline; earned value mechanics and the distinction between physical progress, certified value and recognised revenue; forecast method selection and defence; trend and change control; contingency governance; the habit of auditability; and the professional behaviours that make a number believable.

Does not transfer: your sense of what a number should look like. Every experienced practitioner runs a continuous plausibility check against remembered magnitudes — a productivity rate, a proportion of cost that is usually labour, a normal amount of scope growth in detailed design, a typical procurement lead time. That check is what makes you fast, and after a sector move it is actively dangerous, because it is now calibrated to a different physics and a different commercial structure. §8 works an example of exactly this failure.

The practical response is to treat your own benchmarks as suspect for a full cycle, and to rebuild them from the new project's own data at the first opportunity: measure the achieved rate yourself rather than accepting either your old number or the new team's assertion.

3. Engineering, procurement and construction

Engineering, procurement and construction (EPC) is treated first because it contains the other sectors' problems in one contract.

Contract and risk. Typically lump-sum or turnkey, with the contractor carrying interface risk across three phases that behave completely differently.

Unit of production. Three units in one project: engineering measured in deliverables and their revisions; procurement measured in order placement, fabrication and delivery milestones; construction measured in installed quantities. The integrated progress figure is a weighted composite, and the weighting is a decision that materially changes the reported percentage.

Cost shape. Front-loaded engineering, a very large procurement middle, and a labour-intensive construction and commissioning tail.

Assurance. Client-side project management team, often with a technical adviser; heavy documentary requirement around vendor data and certification.

What transfers in easily. Anyone from a construction-only background brings the installed-quantity discipline. Anyone from procurement brings commitment control.

What must be relearned. Engineering progress. Deliverable-based credit is easy to inflate — a drawing issued for review is not a drawing approved — and engineering slippage is the leading indicator of everything that follows.

Characteristic incomer error. Reading procurement commitment as progress. Placing an order is not achievement; it is exposure. See `BPG-18 — Interface and subcontractor controls` for the interface mechanics.

4. Oil and gas

Contract and risk. Wide range, from reimbursable front-end engineering through to lump-sum construction; heavy long-lead equipment content; turnarounds and shutdowns run to fixed windows with severe consequences for overrun.

Unit of production. Conventional construction quantities in greenfield work; in a turnaround, the unit is the job card and the constraint is the window.

Cost shape. Equipment and long-lead items dominate capital projects. In a turnaround the cost is labour and the schedule is the money.

Assurance. Safety and permit regimes that directly constrain productive hours; extensive technical assurance and, on operating assets, production-loss economics that dwarf the project cost.

What transfers in easily. Cost discipline and procurement control transfer well from any capital project background.

What must be relearned. The productive-hour fraction under a permit regime, worked through in §8; and the cadence of a turnaround, which is not monthly. A shutdown that reports monthly has reported once.

Characteristic incomer error. Applying a monthly controls cycle to a fixed-window outage. In that environment the cycle is daily or by shift, the forecast is remade continuously, and the only question that matters is whether the window will be met.

5. Rail and infrastructure

Contract and risk. Frequently public or regulated clients, with stage-gate funding, defined approval points and formal change governance. Risk allocation is often explicit and heavily documented.

Unit of production. Physical quantities, but gated by access. The controlling unit is frequently the possession or access window rather than the quantity itself.

Cost shape. Civil-heavy, with significant third-party costs — utilities, consents, land, protective works — that are outside your organisation's control and inside your forecast.

Assurance. The heaviest of the six. Public funding brings stage gates, external review and an expectation that every position can be evidenced long after the fact.

What transfers in easily. Documentation discipline from any regulated environment; risk quantification skill, which is used seriously here.

What must be relearned. Access as the primary constraint, third-party dependency management, and float ownership. In an access-constrained programme, float is not spare time; it is the buffer protecting a window that cannot move, and treating it as free is how a recoverable position becomes an unrecoverable one.

Characteristic incomer error. Forecasting third-party dependencies with the same confidence as self-performed work, and treating consent processes as durations rather than as events with their own governance.

6. Data centre

Contract and risk. Frequently a repeatable, modular scope delivered to a client whose commercial value is concentrated in a single date — energisation and handover of capacity. Change late in the programme is common because the end user's requirements evolve.

Unit of production. Repeatable units early — halls, modules, racks of infrastructure — and commissioning systems late. Commissioning is a discipline with its own progressive levels of test, and it is where the programme is actually won or lost.

Cost shape. Heavy mechanical and electrical plant content, with long-lead equipment driving the programme, and an installation labour component that is intense and compressed.

Assurance. Client technical assurance and a rigorous witnessed-test regime; extensive documentation as a condition of handover.

What transfers in easily. Any background with repeatable-unit production; strong procurement and expediting skill, which matters more here than almost anywhere.

What must be relearned. Commissioning as a controlled programme with its own rules of credit, not as a tail activity assigned a percentage. Also the primacy of the date: on a project where the date carries the commercial value, a cost-optimal decision that risks the date is not a saving.

Characteristic incomer error. Crediting commissioning progress by elapsed effort rather than by systems accepted, so the programme reports comfort until the witnessed tests begin.

7. Power, renewables, and building

Two further sectors, treated more briefly because their structures are widely understood.

Power and renewables. The work is packaged — civil works, plant supply, installation, and grid connection — and the packages interlock at dates rather than continuously. Two features dominate the controls job. First, grid connection is an external dependency you do not control and cannot expedite, which makes it a risk position rather than a schedule activity. Second, installation is weather-constrained: forecasting on an average rate rather than on the mechanics of workable windows will be optimistic every time, and the error compounds as the season closes. Payment is usually tied to performance testing and take-over, so the commercial position is concentrated at the back end where the schedule risk also sits.

Building. Measured work against a bill of quantities, a valuation and payment-application cycle, very high subcontract density, and a design responsibility split that varies by procurement route. The most common incomer confusion is between certified valuation and earned value: the valuation is what the client has agreed to pay this period, which is a commercial negotiation, and the earned value is what performance has earned against budget. They rarely match, they should not be forced to, and the gap between them is itself a management question. `BPG-06 — Progress measurement and rules of credit` treats the distinction properly.

8. How this goes wrong

Carrying benchmarks across the boundary. The single largest source of error, and the subject of §9. A rate that was conservative in one environment can be recklessly optimistic in another.

Assuming the vocabulary means the same thing. Float, progress, commitment, completion and even "handover" carry different operational definitions between sectors. Ask for the definition in use before you report against it; do not assume your own.

Underestimating the assurance overlay. Moving from a lightly assured environment into a heavily regulated one, practitioners often treat documentation as bureaucracy and are then unable to evidence a position they held correctly. In these environments an unevidenced correct answer scores zero.

Overestimating it in the other direction. Moving the other way, some practitioners import a governance load the project cannot absorb and slow the cycle below the decision cadence. Both directions are failures to read the fifth variable.

Deferring to the local team on everything. Sector knowledge is real and worth respecting, but "that is how it is done here" is not an answer to "why is the accrual incomplete". The method is yours; the context is theirs.

Presenting your experience in the old sector's units. A hiring manager cannot evaluate "managed the cost position on a lump-sum EPC package" if they think in systems accepted and energisation dates. Translate the evidence into their unit of production before the conversation, using the structure in `CAR-07 — Building a portfolio of evidence`.

Expecting the market to be described here. This document says nothing about which sectors are growing, hiring or paying, because the Institute has no survey data yet and will not assert what it cannot evidence. See `SAL-01` for the instrument and `SAL-06` for how to read anyone else's numbers.

9. Worked example — a benchmark that survived the move and should not have

Illustrative figures. Currency-neutral, labour-hours only. A practitioner moves from a building fit-out environment to a permit-controlled operating facility and carries one number with them: a cable installation rate of 3.0 labour-hours per metre, which was reliable in the previous sector. The scope is 2,000 m. Assumptions: a 10-hour shift; a crew of 12; the same crew composition and the same physical installation difficulty in both environments — that is, the only change is the working regime.

The new environment's working regime. Each shift loses 1.5 hours to permit issue, isolation confirmation and gas testing, and 1.0 hour to travel and muster. Productive time per shift = 10 − 1.5 − 1.0 = **7.5 hours**.

The productive-hour fraction. 7.5 ÷ 10 = 0.75, so only 75 % of each paid shift hour is available at the work face. The correction factor is the reciprocal: 10 ÷ 7.5 = **1.333**.

The corrected rate. 3.0 h/m × 1.333 = **4.0 labour-hours per metre**.

The effect on the estimate. At the carried rate: 2,000 m × 3.0 = **6,000 hours**. At the corrected rate: 2,000 m × 4.0 = **8,000 hours**. The carried benchmark understates the labour by **2,000 hours**, which is 2,000 ÷ 6,000 = **33.3 %** of the original estimate.

The effect on the programme. Crew capacity = 12 people × 10 hours = 120 site-hours per shift. At 6,000 hours: 6,000 ÷ 120 = **50 shifts**. At 8,000 hours: 8,000 ÷ 120 = 66.7 → **67 shifts**. The difference is **17 shifts** of duration that were never in the programme, on an activity that was priced and sequenced with confidence because the practitioner had used that rate successfully many times.

What should have transferred. Not the rate — the method. The transferable technique is to measure the productive-hour fraction in the new environment before committing to any rate: observe or reconstruct one shift, subtract the regime losses, and derive the factor from the project's own data. Then verify it against timesheets in the first full cycle and correct it. The professional who does this in their first month is applying exactly the same skill that made their old benchmark reliable, which is the point: the skill transferred, the number did not.

10. Checklist — making a sector move

- Answer the five questions in §1 about your new sector, in writing, before your first reporting cycle.
- List every benchmark you rely on — rates, percentages, lead times, typical growth — and mark each as unverified until you have derived it from this project's data.
- Get the definitions: what counts as started, complete, committed, handed over, and what float means contractually here.
- Find out who audits you, how often, and what they ask for. Read the last report they issued.
- Establish the decision cadence and design the cycle to serve it, not the cycle you are used to.
- Identify the single dominant constraint — access window, long-lead equipment, weather, permit regime, energisation date — and check that your reporting makes its status visible every period.
- Translate your prior evidence into the new sector's unit of production before your first interview or first internal presentation.
- Plan the entry itself with `CAR-08 — The first ninety days in a controls role`.

The practitioners who move sectors well are not the ones who knew the new sector already. They are the ones who arrived assuming their instincts were wrong, and spent the first cycle rebuilding them from the new project's own data — which is why the move costs a cycle rather than a career.

Related

- `CAR-01 — The project controls career roadmap` — the six disciplines that transfer intact
- `CAR-02 — Routes into project controls` — the entry routes, whose gaps behave similarly to sector gaps
- `CAR-08 — The first ninety days in a controls role` — the plan for the first cycle after a move
- `BPG-06 — Progress measurement and rules of credit` — the distinction misread in §7
- `BPG-18 — Interface and subcontractor controls` — the interface problem that dominates EPC work
- `SAL-03 — Role taxonomy and levelling` — how roles are compared across sectors
- `SAL-06 — Using market data honestly` — why this document contains no sector market claims

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 5 (cost management and cost control), Domain 6 (earned value management and forecasting), Domain 7 (contracts, commercial management, bills of quantities, invoicing and revenue) and Domain 10 (project scheduling). Sector characterisations are structural descriptions of contract forms, production units and assurance regimes; no named project, organisation, market statistic or demand claim is used, and none should be inferred.

Status and version

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CAR-07 · CAREER ROADMAP

Building a portfolio of evidence

What evidence of controls work actually looks like, and how to describe it without breaching confidence

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

Summary

A curriculum vitae asserts; a portfolio demonstrates. This document gives a five-part structure for an evidence record, a graded scale from presence to ownership, an inventory of where evidence appears in an ordinary reporting month, and — the part most people get wrong — a method for anonymising a case properly, including why anonymisation fails on combinations of details rather than on names, and how to index figures so the teaching survives while the confidential magnitudes do not.

About this document

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Contents

Building a portfolio of evidence

In one paragraph. A curriculum vitae asserts what you did; a portfolio demonstrates what you decided, what changed because of it, and how you know. This document gives a five-part structure for an evidence record, a scale that distinguishes presence from ownership, an inventory of the evidence an ordinary reporting month already produces, and a seven-step method for anonymising a case — including the failure that catches most people, which is that anonymisation is defeated by a combination of ordinary details rather than by a name. It ends with a before-and-after anonymisation, worked with arithmetic, showing that the ratios carrying the teaching survive indexation while the confidential magnitudes do not.

Who this is for. Cost engineers, planners, risk and change professionals and controls managers building a record of their own work; also hiring managers and assessors deciding what to ask for.

1. Why a portfolio, not a longer CV

A curriculum vitae lists positions occupied. It answers "where were you" and, by implication, "for how long" — and neither question separates two people with the same title on the same project, one of whom owned the forecast and one of whom formatted the report.

An interviewer, a competency assessor and a client's technical panel want three other things. **What did you decide** — not what the team decided; a decision has an owner, a date and a rejected alternative. **What changed because of it** — a number that moved, an error prevented, a claim defended, a control that stopped a class of failure rather than one instance. **How do you know** — evidence someone else can test, including the record of the times you were wrong. A portfolio is those three questions, answered repeatedly, in a structure someone can read quickly.

2. The unit of evidence

One record, five parts. Fitting on one page is a feature.

1. The claim. One sentence naming the competency and the level. "Owned the forecast at completion for a package of control accounts, including method selection and defence to the project director."

2. The context. Enough to make the claim assessable and no more: type of project, contract form, relative scale, your role and reporting line. Not the client, project or contract value — §5 and §6 explain why.

3. The action. What you personally did, in the first person singular, with the decision explicit and the alternative named. "I selected a performance-persists forecast rather than the one-off-variance basis used the previous cycle, because the shortfall was present in three consecutive cycles and in both the timesheet and the quantity record."

4. The outcome. What changed, quantified where it can be quantified without disclosure — proportions, ratios and index points rather than currency. Include outcomes that went against you; they are more persuasive, not less.

5. The verification. How a third party could check: a referee by role, a countersignature, an audit reference, an artefact that exists independently of you. §7 covers this.

A record missing part 3 is a job description; a record missing part 5 is an assertion. Both are common and both are discounted immediately by anyone who reads portfolios for a living.

3. The scale that matters: presence, contribution, ownership, change

Weak evidence is usually weak because it sits at the wrong rung of this scale.

Level	What it sounds like	What it evidences
Presence	"Worked on the monthly reporting cycle"	That you were there
Contribution	"Supported the preparation of the forecast"	That you were useful
Ownership	"Owned the forecast for eleven control accounts, including method selection, and defended it at the monthly review"	That you can be held to an answer
Change	"Found that the accrual routine was missing subcontract work certified after cut-off; redesigned it; the correction moved reported cost performance and stopped the understatement recurring"	That the organisation is different because you were in it

Ownership and change are what get read, and neither can be manufactured when you need it — only captured while it happens. `CMP-02 — Competency levels and how they are evidenced` sets out how the Institute levels competence; this is about the record you keep for yourself.

4. Where the evidence already is

An ordinary reporting month generates more evidence than most people record in a year. The inventory:

- the accrual position you established at cut-off, and the judgement you made about an incomplete input;
- the variance narrative you wrote, and the driver you identified before it reached the ledger;
- the change you priced, with the basis, and what happened to it;
- the schedule update narrative and the delta explanation (`CAR-04` §6);
- the risk you added that later materialised, or the one you argued to retire;
- the audit finding you responded to, and the control you changed as a result;
- the recovery plan you tested for plausibility, and the reconciliation you ran between two systems that disagreed;
- the template, mapping, rule of credit or automation you built that outlived your involvement;
- the error in your own work that you reported before anyone else found it;
- the forecast you published in advance with a falsification condition (`CAR-03` §6) and how it behaved.

Ten minutes at the end of each cycle, capturing one record from that list, produces a portfolio with nothing reconstructed after the fact — and reconstruction is where accuracy and confidentiality both fail.

5. Confidentiality: what you may not take, and the test to apply

The governing principle is unwelcome and simple: **your employer's and your client's data is not your portfolio.** Yours is your own account of what you decided and why. Not yours: documents, extracts, screenshots, datasets and models, whether or not you personally created them. Specifically, do not disclose:

- the identity of the client, project or contract, unless it is genuinely public and your employer is content; and any value, rate or margin, or anything from which one can be derived;
- unpublished claims, disputes, notices, entitlement positions, or subcontractor pricing;
- personal data about colleagues, including performance, disciplinary and safety matters;
- security-sensitive site, system or asset information, which in some sectors is a legal matter;
- anything under a non-disclosure agreement, and in listed-company contexts anything price-sensitive.

The three-reader test. Before a record leaves your possession, ask whether it could be read without harm or embarrassment by the client, by your employer's commercial director, and by a competitor. If any answer is no, it is not ready. That is stricter than "would anyone recognise it", and it is the right test, because harm from disclosure does not require recognition.

Where a conflict is live between what you may say and what you believe should be said, the Institute's treatment is in `ETH-04 — Conflicts of interest in project controls`.

6. Anonymising a case properly

Most attempted anonymisation removes the name and stops, which is not anonymisation but redaction of the single least informative field. Work through all seven steps.

Step 1 — remove direct identifiers. Client, project, contract number, site, joint-venture partners, named individuals. Replace people with roles: "the project director", "the client's quantity surveyor".

Step 2 — remove the identifying combination. This is the step that is missed. Sector, region, approximate value, contract form and period, taken together, identify a project to anyone in that market even though no single element names it. The test is not "have I removed the name" but "how many projects in the world match this description". If the answer is a small number, keep generalising: widen the sector descriptor, drop the region, express the period relatively rather than by year.

Step 3 — index the figures. Do not round confidential magnitudes; remove them. Choose a base — the budget at completion is usually the natural one — set it to 100, and express every other figure as a proportion of it. State plainly that figures are indexed. §10 works this through.

Step 4 — keep the ratios, because they carry the teaching. Performance indices, variance percentages, proportions of budget, days of movement and productivity factors all survive indexation. A reader learns nothing from a currency amount they cannot contextualise and everything from an index that moved from 1.09 to 1.00.

Step 5 — check for re-identification through the incident itself. A distinctive event — an unusual failure, a well-known dispute, a notable delay — can identify a project more precisely than any figure. Describe its mechanism without the case, or do not use it.

Step 6 — label the anonymisation. In the record: "Figures indexed to a budget of 100. Project, client and period not identified. No confidential document is reproduced and no figure can be derived from this record." A reader who can see the discipline you applied trusts the rest more.

Step 7 — seek permission where it is possible, and record the answer. For anything more detailed than an indexed record, ask in writing and keep the reply. If refused, record the refusal and use the method-only version. Silence is not consent.

Where even the indexed case is identifiable, write the method instead. "How to test whether a recovery claim is plausible" teaches what the case would have taught, cites no project, and can be shown to anyone.

7. Verification without disclosure

Evidence that cannot be checked is an assertion with extra sentences. Three forms disclose nothing: **a referee by role** ("the project controls manager to whom I reported can confirm this", the name supplied privately when needed, never published); **a countersignature**, a line manager signing a competency record, which is what structured development schemes rely on and what `CMP-10 — Assessing competence` describes for formal assessment; and **an independent artefact or audit trace**, such as a procedure you wrote that stayed in use, or "this control was tested in an internal audit and the finding closed", referenced by type and period rather than by number.

Assessed knowledge is the fourth, and it verifies something different. A credential evidences knowledge against a published standard, examined independently of your employer. PCI's project controls credential is PCL-AI; it is examined, not portfolio-assessed, and PCI is developed with reference to ISO/IEC 17024 personnel-certification principles and is not accredited. Credential and portfolio evidence knowledge and practice respectively, and neither substitutes for the other. A membership grade is a third thing again — a standing, not a certification — and conflating the ladders looks careless.

Continuing professional development records verify differently again, being structured: PCI recertification runs on a three-year cycle with a mandatory AI-currency component, and only approved entries in the current cycle count. The binding volume is [CONFIRM: required CPD hours per three-year cycle — the student portal currently shows a 30-hour target, and the binding requirement will be published with the recertification rules]; `CER-08` owns the rules.

8. Structuring the portfolio itself

Organise by competency, not by employer: an employer-ordered portfolio makes the reader do the mapping, and readers do not. Use the PCL-AI competency set as the index: project controls governance, cost management, planning and scheduling, earned value management, forecasting and estimate at completion, performance measurement, project risk, commercial and contract controls, predictive analytics, AI-enabled project controls, digital reporting, automation, responsible AI, and human validation.

One page each, carrying two records at the highest level you can honestly claim and one line naming the gap. That gap line is the most credible thing on the page, because a portfolio with no gaps reads as one its author has never audited.

Keep two versions and only two: a full one, with names, for your own reference and for a conversation under non-disclosure, and a shareable one anonymised by §6. A third, half-anonymised version is how a name eventually escapes. On leaving an employer, capture your own record of decisions and take nothing else — no documents, extracts, data or models. That is a legal boundary in most jurisdictions, and the part you may keep is the only part a future employer values.

9. How this goes wrong

Writing it at the end. Portfolios assembled during a job search are vague about the details that matter — the date, the alternative rejected, the number before and after — because those were never captured and cannot now be reconstructed honestly.

Claiming the team's work. "We reduced the forecast" evidences nothing about you. Assessors probe collective verbs first.

Removing the name and calling it anonymised. Step 2 of §6 is the step that gets skipped, and the result is a portfolio that identifies a client project to every reader in the sector while its author believes it is discreet. Its close relative is rounding rather than indexing: a rounded budget still discloses the magnitude, which is usually the confidential part.

Including a document. A report screenshot, a register page, a model extract — all the employer's property regardless of who typed them, and any one of them converts a professional record into a disclosure incident.

Only recording successes. Unbroken correctness is not believed, because nobody who forecasts has that record. A documented wrong forecast, with what you learned and changed, is stronger evidence of judgement than three correct ones — and a list of software proficiencies weaker than either.

10. Worked example — one record, before and after anonymisation

Illustrative figures. Currency-neutral, one control period. Assumptions: progress measured to agreed rules of credit; the accrual identified was the complete un-invoiced position at cut-off; no change approved in the period.

The confidential version — accurate, and unpublishable.

> On the [named client, named project], budget at completion 42,000,000, I identified at the October cut-off that subcontract liabilities of 1,260,000 had not been accrued. Earned value at that date was 15,000,000 against invoiced cost of 13,800,000.

The arithmetic. Cost performance index before correction = earned value ÷ actual cost = 15,000,000 ÷ 13,800,000 = **1.087**. Actual cost with the accrual booked = 13,800,000 + 1,260,000

= **15,060,000**. Index after = 15,000,000 ÷ 15,060,000 = **0.996**. One entry moved reported performance from 8.7 % above parity — work earned exceeding cost incurred — to marginally below it.

The anonymised version — publishable, and it teaches the same lesson.

> On a lump-sum industrial project, in its construction phase, I owned the accrual position for a package of control accounts. At a month-end cut-off I identified that subcontract liabilities equal to **3.0 index points** of the budget had not been accrued. Earned value at that date stood at **35.7 index points** against invoiced cost of **32.9**. Booking the accrual moved the reported cost performance index from **1.085 to 0.994** — from apparently earning more than it was spending to marginally the reverse — on a single entry. I subsequently redesigned the cut-off routine so that subcontract work certified after the invoicing deadline is captured by default. Figures indexed to a budget at completion of 100; project, client and period not identified; no confidential document is reproduced.

Checking the indexation. Base = budget at completion = 42,000,000 = 100 index points. Earned value: 15,000,000 ÷ 42,000,000 × 100 = **35.7**. Invoiced cost: 13,800,000 ÷ 42,000,000 × 100 = **32.9**. Accrual: 1,260,000 ÷ 42,000,000 × 100 = **3.0**. Index before: 35.7 ÷ 32.9 = **1.085**, against 1.087 computed on the raw figures. Index after: 35.7 ÷ (32.9 + 3.0) = 35.7 ÷ 35.9 = **0.994**, against 0.996 on the raw figures.

The drift of two thousandths in each index is rounding introduced by publishing indexed values to one decimal place, and it changes nothing a reader would conclude. That is the test of a good anonymisation: the magnitudes are gone and unrecoverable, the ratios are intact to within stated rounding, and the lesson — that one missed accrual can move a position across the line between earning and overspending — is as instructive as in the confidential version.

What else went, and why. The budget, as the identifying magnitude; the named month, because a cut-off calendar is a fingerprint; and the specific process industry, widened to "industrial", per step 2.

11. Checklist — the portfolio audit

- Every record has all five parts in §2, is written in the first person singular, and names the decision and the alternative rejected.
- At least half your records sit at ownership or change on the scale in §3, and at least one describes something you got wrong and what changed as a result.
- No record contains a client name, project name, contract value, rate, margin or extract of a document.
- Every case has been through all seven anonymisation steps, not only step 1, and every figure is indexed to a stated base with the indexation labelled.
- You have applied the three-reader test in §5 to the shareable version.
- Permissions sought are recorded, including refusals.
- The portfolio is indexed by competency, with a stated gap on each page.
- You captured at least one record in the last reporting cycle.

The cost of skipping it is paid later and in full: by the time you need the evidence, the project has finished and the people who could verify it have moved, leaving a job title and a date — which is exactly what everyone else in the room also has.

Related

- `CAR-01 – The project controls career roadmap` — the transitions this evidence demonstrates
- `CAR-03 – Cost engineer to cost manager` — the forecast record described in §4
- `CAR-05 – Into project controls leadership` — movement bridges as leadership evidence
- `CMP-02 – Competency levels and how they are evidenced` — how the Institute levels competence
- `CMP-10 – Assessing competence: evidence, rubrics, moderation` — formal assessment, not a personal portfolio
- `CER-08 – Recertification, CPD and the AI-currency requirement` — the rules behind the §7 placeholder
- `ETH-04 – Conflicts of interest in project controls` — when confidentiality becomes conduct

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domain 4 (performance management, variance analysis and management reporting), Domain 5 (cost management and cost control) and Domain 6 (earned value management and forecasting); and from the PCL-AI competency set, used as the portfolio index in §8. Confidentiality guidance is stated as principle, not as the law of any jurisdiction: disclosure obligations, employment terms and data-protection rules vary, and a specific case must be checked against the applicable contract and law. No external market, salary or demand data has been used.

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CAR-08 · CAREER ROADMAP

The first ninety days in a controls role

What to read, whose trust to earn, and which numbers to verify personally before you inherit them

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

Summary

On day one in a controls role you become accountable for numbers you did not produce and cannot yet check. This is a playbook for converting inherited assertions into verified positions before your name goes on a report: what to read and what to extract from each document, the ten numbers to verify personally with the test for each, whose trust to earn and in what order, what not to change in the first cycle, and the reconciliation that must tie before anything else is worth doing.

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Examples. All worked examples are illustrative and fictional. Figures are chosen to demonstrate a method, not to describe any real project, organisation or market. Sample examination items, where they appear, are study material maintained separately from any live examination bank.

Completeness. Passages marked [CONFIRM: ...] are operational or market facts that have not yet been confirmed from a cited source. A document carrying them is a working draft and is not approved for publication.

Contents

The first ninety days in a controls role

In one paragraph. A new controls role hands you accountability for numbers you did not produce, that were derived by people you have not met, from a process you have not seen. The first period is therefore not about impressing anyone; it is about verification, and about earning the access that makes verification possible. This document sets out what to read and what to extract from each item, ten numbers to check personally with the test for each, whose trust to earn and in what order, what not to change in the first cycle, and what must be true before you sign your first report. Its structure is the reporting cycle, not the calendar week, because the cycle is the clock a controls role actually runs on.

Who this is for. Anyone starting in a cost, planning, risk or controls management role — a first role, a new employer, a new project, or a sector move — and the managers responsible for their handover.

1. The inheritance problem

On your first day the project already has a budget, a baseline, a forecast, a risk provision and a reported position — all produced by someone else, much of it by someone who has left, and within a cycle or two it will be described as yours. From then on every error in it is your error, including those that predate you.

Hence the characteristic failure of a new controls professional: inherit the position silently, discover the problem later, and then have to explain both the problem and why you reported it as sound twice. The alternative is not to distrust everything loudly. It is to verify a specific, short list of things quickly, in a way indistinguishable from ordinary diligence, and to say what you found before you have signed anything.

Organise it in three phases by reporting cycle rather than by week, because how much calendar time you get before your first cut-off is an accident of your start date.

Phase 1 — before your first cut-off. Read, meet, and reconcile the budget. Nothing else. **Phase 2 — your first full cycle.** Run the cycle as it is run, verify the ten numbers in §3, and record what you find without changing anything. **Phase 3 — your second cycle.** Publish a position you have verified, and change exactly one thing.

2. What to read, in order, and what to take from each

The order matters: each item makes the next one intelligible.

The contract. Not a summary of it. Extract, onto one page: what triggers payment and on what evidence; what constitutes a change and who may instruct one; the notice provisions and their time bars; the programme obligations; any liquidated damages or bonus mechanism; and the definitions clause, because contracts routinely define "cost", "completion" and "programme" in ways that differ from ordinary use. That page will be consulted more often than anything else you produce.

The project execution plan and the controls procedure. What the function has committed to do, to whom, by when. The gap between that and what is actually happening is your first diagnostic and frequently your first improvement.

The breakdown structures and their mappings. The work breakdown structure, the cost breakdown structure, the code of accounts, and how each maps to the enterprise resource planning system and to the contract's payment structure. Ask for the mapping document; if none exists, that is a finding, and it explains most of what you will later discover about reconciliation.

The current baseline and the approved change log. Which baseline is current, when it was approved, and every change since. `BPG-04 — Baselining and baseline change control` treats the discipline; what you need on arrival is "what am I measuring against, and who agreed it".

The last three monthly reports, read backwards. Looking for two things: numbers that moved without explanation, and narratives that repeat unchanged across all three. A repeated narrative means either a genuinely static issue or a text box nobody updates, and the second is far more common.

The risk register and the last two review minutes. Whether risks are being retired, whether anything has been added recently, and whether the quantification bears any relationship to the contingency held.

The trend and change register, the pending items especially: their total value, their age, and what happens to the forecast if every one is refused. Then **the last audit or assurance report** — somebody has already looked, so read what they found, what was promised, and whether it happened — and **the cash-flow forecast against actual receipts**, to see whether it is built on certification or invoice dates and how the two have diverged.

The organisation chart and the delegated authority matrix. Who may commit money, up to what value, and who approves change — read against the approval trail on the last few change orders to check they agree.

3. The ten numbers to verify personally

Not audited — verified. Each can be checked in hours, by you, before you have any authority at all. Each is stated as the question, the test, and what a bad answer sounds like.

1. Does the budget reconcile? Test: control-account budgets plus undistributed budget plus management reserve equals the approved project budget, and that equals the contract or sanction value plus approved changes. §9 works it through. Bad answer: "it ties in the system." It either reconciles on a page you can show someone, or it does not.

2. Does actual cost reconcile to the ledger? Test: the cost report's actual cost for a stated period against the general ledger for the same period, differences named. Bad answer: an assurance that finance and controls "use the same system".

3. Do commitments reconcile to open orders? Test: the commitment register against open purchase orders and subcontracts in the source system. Bad answer: a register maintained manually in parallel, which is where cancelled and duplicated commitments accumulate.

4. Is the accrual complete? Test: take the three largest subcontracts and compare the last accrual against certified or measured work at the same date. Bad answer: the same round number every period.

5. Is earned value derived or asserted? Test: pick three control accounts and re-derive earned value from the rules of credit and the raw quantities. Bad answer: a percentage supplied by the delivery team and entered without a rule.

6. What is actually driving the completion date? Test: run the driving path yourself; check constraints, open ends, out-of-sequence progress and the data date. Bad answer: a critical path running through administrative activities, or one that changes character every update.

7. What is left in contingency, and is any of it spoken for? Test: the drawdown record — what was held, what released, on whose authority, against which risk. Bad answer: a figure with no drawdown history.

8. What is the exposure if every pending change is refused? Test: total unapproved change and its treatment in the forecast. Bad answer: pending change carried as though approved, with no separate line.

9. Is anything counted twice? Test: compare the risk provision against the forecast for events funded in both — the pattern in `CAR-01` §7. Bad answer: nobody has ever compared them.

10. Is the cash forecast built on the right dates? Test: whether receipts are forecast from certification, invoice or payment dates, and what the historical lag has been. Bad answer: the cost forecast with a fixed delay applied.

Write down what you find, dated, as you find it — not as an accusation but as a handover record. If the position is sound, you now have evidence that it was sound when you arrived, which is worth as much as finding a problem.

4. Whose trust to earn, in what order

Access makes verification possible, and access is granted socially before it is granted formally. Arrive with a question, not an opinion.

The project director or project manager. First, and within days. Agree the escalation contract described in `CAR-05` §5. Ask what they do not trust in the current reporting; the answer is usually specific and is the most useful thing you will hear in the first month.

The commercial or quantity surveying lead. They own the boundary of what may be claimed. Ask what is contested, what notices are live, and what they need from controls that they are not getting.

The finance controller or project accountant. They own the ledger you must reconcile to. Ask for the last reconciliation between the cost report and the ledger, and for the cut-off calendar finance actually works to, which is not always the one controls assumes.

Your counterpart in the other discipline — the planner if you are cost, the cost engineer if you are planning. Agree how the two positions will be reconciled each period and who resolves a disagreement.

The delivery and construction managers. They own the truth you are reporting. Go and look at the work, early, so that a reported percentage means something physical to you. This one habit does more for a controls professional's credibility than any analysis.

Procurement and expediting, for commitment, lead times and delivery dates, which drive the cost position and, on equipment-led projects, the programme; **document control**, who hold the evidence and know which registers are actually maintained, where an hour is worth a week of searching; and the **subcontract administrators**, where the cost risk lives and accrual accuracy is decided.

5. What not to do in the first cycle

Do not rebaseline. Whatever is wrong with the baseline, resetting it in your first cycle destroys the variance history that would have explained the project to you, and it looks exactly like a new arrival hiding an inherited problem.

Do not rebuild the report pack, and do not change the tool. You do not yet know which oddities are habits and which are contractual requirements, and every system migration proposed in a first month is a proposal to spend the project's attention on your comfort. Redesign belongs in phase 3 at the earliest.

Do not correct your predecessor in public. State findings impersonally and forward: the position, the effect, the fix. The person you criticise usually has friends who will be reviewing your numbers.

Do not promise a recovery you have not tested. Apply the plausibility test in `CAR-04` §9 before showing any recovery in any document with your name on it.

The one exception. If you find something material and wrong — material meaning it would change a decision — say so immediately, quietly, upward, with the evidence and your proposed treatment. Raising it early costs a difficult conversation; raising it late costs your credibility, permanently.

6. Your first report

Before you publish a report with your name on it, four things must be true.

The budget reconciles, on a page, per §3 item 1 and §9. **Actual cost reconciles to the ledger** for the period, with differences named. **You can trace at least one number end to end** — from a timesheet or measured quantity to the reported figure — and you know every hand-off it passed through. **Every judgement is labelled as one.** Anything accepted without verification is stated as inherited: "the risk provision is carried forward and has not yet been re-quantified" is an honest sentence that costs nothing and protects everything.

Include a movement bridge from the previous period (`CAR-05` §8) even if the movement is small: introducing one now is easy, whereas introducing one in a month when the number has moved badly looks defensive. Add a short "known unknowns" note — what you have not yet verified, and when you will have — which converts every subsequent discovery from a failure into a completed task.

7. The one thing to change

By your second cycle you will have a list. Change one thing, and choose it by a single criterion: which control would have caught the last unpleasant surprise on this project?

The answer is usually one of a small set — an incomplete accrual routine, a rule of credit that permits progress without evidence, a mapping that forces manual reconciliation, a cut-off nobody enforces, or a pending-change register absent from the forecast. Fixing one visibly, and showing what it caught, buys the authority for the next five. Attempting all five at once usually delivers none.

`BPG-19 — Project controls assurance and health checks` and `TPL-15 — Project controls health check` give the structured diagnostic, if you would rather assess the whole function than pick by instinct.

8. How this goes wrong

Signing the inherited position. The single most damaging first-cycle decision: publishing a report you have not verified, because it was the report the project already produced. From that point the position is yours retrospectively.

Verifying everything and reporting nothing. The opposite failure. A new arrival who spends three cycles auditing and misses two reporting deadlines has demonstrated the wrong competence.

Arriving with the last project's answers. Particularly after a sector move — see `CAR-06` §2. The method transfers; the benchmarks do not, and importing them makes you confidently wrong in front of people who know better.

Mistaking politeness for agreement. "We will look into that" is not a commitment. Anything that matters is confirmed in writing, in the ordinary record, the same day.

Confusing access with trust. Being given the files is not the same as being told what the files do not say, and the second only arrives once you have shown you will not use it against the person who told you.

Escalating too early with too little. A finding raised without a quantified effect and a proposed treatment is a complaint. Take the twenty minutes to quantify it; it changes how it is received entirely.

Letting the handover lapse. If your predecessor is still available, the questions worth asking are not about systems: they are which numbers they were least comfortable with and what they would have fixed next.

9. Worked example — the budget reconciliation that does not tie

Illustrative figures. Currency-neutral units, one project, at the data date of your first cut-off. Assumptions: management reserve is held outside control accounts; all variations listed are formally approved; earned value is measured to agreed rules of credit against distributed budgets.

The three statements you are given. Original contract value: 38,400,000. Approved variations to date: 1,150,000. Approved project budget therefore = 38,400,000 + 1,150,000 = **39,550,000**.

Sum of control-account budgets in the cost system: 39,010,000. Management reserve held: 400,000.

The reconciliation. Distributed plus reserve = 39,010,000 + 400,000 = **39,410,000**. Against the approved budget of 39,550,000, the gap is 39,550,000 − 39,410,000 = **140,000** — undistributed and unexplained.

The cause, once you look. One approved variation of 140,000 was added to the approved budget but never distributed to a control account. The scope is being executed and its cost is being charged, but no budget exists against which to earn it.

The effect on the reported position. Suppose 60 % of that variation's scope is complete. Earned value is understated by 0.60 × 140,000 = **84,000**. If the project's reported earned value is 14,200,000 against actual cost of 13,900,000, then:

Reported cost performance index = 14,200,000 ÷ 13,900,000 = **1.022**. Corrected earned value = 14,200,000 + 84,000 = **14,284,000**. Corrected index = 14,284,000 ÷ 13,900,000 = **1.028**.

Six thousandths of an index point, from one undistributed variation. On this project it happens to understate performance, which is the benign direction; the same error with the cost charged and the scope not yet started overstates it, which is not.

The second check, which is the one that matters. Is that same 140,000 also sitting in the pending change register? A variation that was approved, added to the budget, never distributed and left in pending change is carried twice in the forecast — and it will happen to every variation processed the same way, because the defect is in the routine, not in the transaction.

What you do with it. Not an accusation: a finding. The reconciliation, the cause, the effect on the index, the double-count check, and the treatment — distribute the budget, confirm the pending register, and add the reconciliation to the standing cut-off checklist so it is proved every period rather than discovered by the next new arrival.

10. Checklist — the ninety-day plan on one page

Before your first cut-off - Read the contract and produce the one-page extract in §2. - Obtain the breakdown structures and their mapping document, or record that none exists. - Read the last three monthly reports backwards and list every unexplained movement. - Meet the project director and agree the escalation contract; then commercial, finance, your discipline counterpart and document control. - Reconcile the budget per §9 and write the result down, whichever way it comes out.

Your first full cycle - Run the cycle as it is currently run; change nothing. - Verify the ten numbers in §3 and record each finding with its date and test. - Walk the work and compare what you see with what is being reported. - Raise anything material immediately, quietly, upward, with a quantified effect.

Your second cycle - Publish a position you have verified, with judgements labelled, a movement bridge attached and a "known unknowns" note listing what remains unverified and by when. - Change one control — the one that would have caught the last surprise — and show what it caught. - Capture two evidence records for your portfolio, per `CAR-07` §2, while the detail is still exact.

Ninety days is long enough to verify an inherited position and short enough that nobody yet assumes you built it. That window closes quietly and without announcement, and it does not reopen: after it, the numbers are yours whether or not you ever checked them.

Related

- `CAR-02 — Routes into project controls` — the gap you arrive with, which shapes what to check first
- `CAR-05 — Into project controls leadership` — the escalation contract and the movement bridge
- `CAR-06 — Moving between sectors` — extra first-cycle work when the move crosses a sector boundary
- `CAR-07 — Building a portfolio of evidence` — capturing findings while the detail is exact
- `BPG-04 — Baselining and baseline change control` — the discipline behind §9
- `BPG-19 — Project controls assurance and health checks` — the structured alternative to §7
- `TPL-15 — Project controls health check` — the instrument for that diagnostic

Sources and standards

Drawn from the PCI Body of Knowledge, principally Domains 4 (performance management, variance analysis and management reporting), 5 (cost management and cost control), 6 (earned value management and forecasting), 7 (contracts, commercial management, invoicing and revenue), 10 (project scheduling) and 12 (risk management for project controls); and from the PCL-AI competency set. All figures in §9 are illustrative and were computed for this document. No external market, salary or demand data has been used.

Status and version

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